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(54) **ELECTRIC SCOOTER**
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(56) **References Cited**
 U.S. PATENT DOCUMENTS

 213,546 A 3/1879 Cornelius
 215,081 A 5/1879 Turnbull
 (Continued)

 FOREIGN PATENT DOCUMENTS

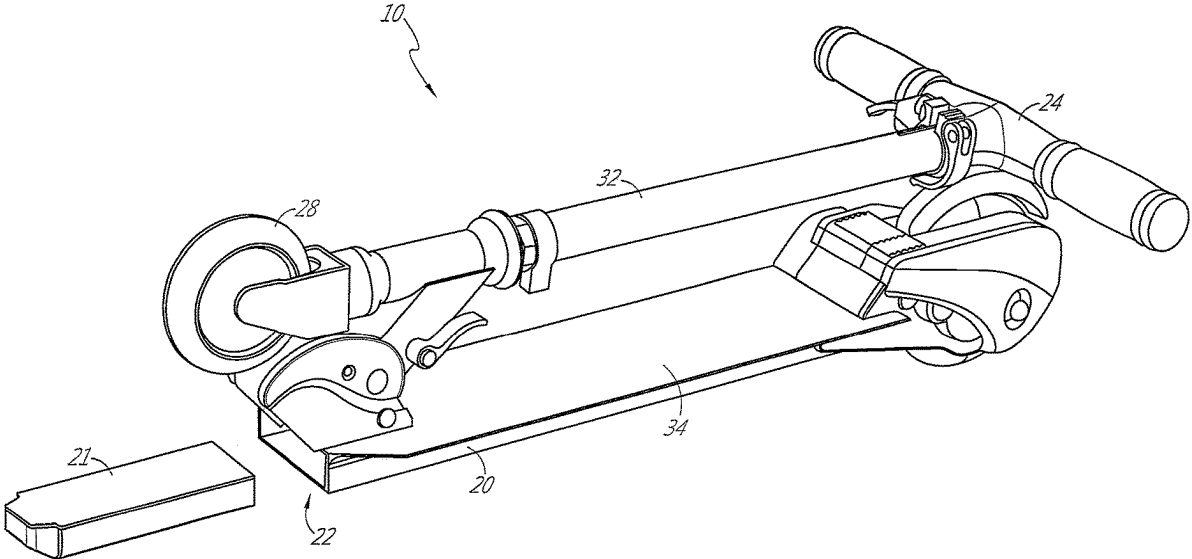
 AU 2005100475 6/2005
 CH 598051 4/1978
 (Continued)

 OTHER PUBLICATIONS

 <https://www.amazon.com/Razor-Turbo-Jetts-Electric-Wheels/dp/BK07H6RB4MX?th=1> (Year: 2018).
 (Continued)

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(57) **ABSTRACT**
Various personal mobility vehicles, such as scooters, are disclosed. The scooter can include at least one battery and motor for powering at least one driven wheel. The vehicle can include a gear assembly to convert a torque produced by the motor to a different torque to a driven shaft to power the at least one driven wheel. The battery can be removable.

22 Claims, 18 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

252,970	A	1/1882	Price et al.	4,058,323	A	11/1977	Ware, Jr.
253,735	A	2/1882	Marshall	4,061,350	A	12/1977	Schmidt, Jr. et al.
304,949	A	9/1884	Mitchell	4,094,372	A	6/1978	Notter
311,936	A	2/1885	Wisewell	4,127,282	A	11/1978	Gorlach
319,839	A	6/1885	Nelson	4,152,001	A	5/1979	Christianson
321,434	A	7/1885	Hanson	4,168,076	A	9/1979	Johnson
321,466	A	7/1885	Wall	4,180,278	A	12/1979	Gottlieb
329,927	A	11/1885	Mendenhall	4,185,847	A	1/1980	Johnson
1,018,512	A	2/1912	Mees	4,272,091	A	6/1981	Ried, Jr.
1,075,250	A	10/1913	Gingold	4,398,735	A	8/1983	Evans et al.
1,194,102	A	8/1916	Webb	4,402,521	A	9/1983	Mongeon
1,387,091	A	8/1921	Woolley et al.	4,403,784	A	9/1983	Gray
1,550,985	A	8/1925	Schluesselburg	4,417,737	A	11/1983	Suroff
1,603,529	A	10/1926	Faust	4,504,072	A	3/1985	Klawitter
1,632,997	A	6/1927	Connolly	4,508,187	A	4/1985	Wenzel
1,728,063	A	9/1929	James	4,523,767	A	6/1985	Le Page
1,784,761	A	12/1930	Smith	4,546,841	A	10/1985	Sipiano
1,854,188	A	4/1932	Gregory	4,645,223	A	2/1987	Grossman
1,935,187	A	11/1933	Kitselman	4,681,333	A	7/1987	Rouge et al.
2,051,762	A	8/1936	Vincent	4,708,352	A	11/1987	Vullierme
2,077,274	A	4/1937	Silkman	4,709,937	A	12/1987	Lin et al.
2,086,471	A	7/1937	Chapman	4,738,456	A	4/1988	Creason
2,126,359	A	8/1938	Vogt	D299,260	S	1/1989	Vullierme
2,136,306	A	11/1938	McCloy	4,842,091	A	6/1989	Badsey
2,216,359	A	10/1940	Spencer	4,844,491	A	7/1989	Wheelwright
2,413,530	A	12/1946	Taylor	4,930,794	A	6/1990	Chan
2,430,037	A	11/1947	Vincent	4,955,626	A	9/1990	Smith et al.
2,460,395	A	2/1949	Reid	4,998,358	A	3/1991	Girardelli
2,474,082	A	6/1949	Wutz	5,046,747	A	9/1991	Nielsen
2,509,324	A	5/1950	Horn	5,062,630	A	11/1991	Nelson
2,857,008	A	10/1958	Antonio	5,092,614	A	3/1992	Malewicz
3,087,739	A	4/1963	Ware	D326,290	S	5/1992	Badsey
3,252,713	A	5/1966	Heller	5,143,388	A	9/1992	Chen
3,288,251	A	11/1966	Sakwa	5,232,235	A	8/1993	Brooks
3,331,612	A	7/1967	Tietge	5,236,058	A	8/1993	Yamet et al.
D214,252	S	5/1969	Andersen	5,263,725	A	11/1993	Gesmer et al.
3,476,399	A	11/1969	Finn	5,330,026	A	7/1994	Hsu et al.
3,570,620	A	3/1971	Fischer et al.	5,347,681	A	9/1994	Wattson et al.
3,671,055	A	6/1972	Aarup	5,513,863	A	5/1996	Klamer et al.
D226,440	S	3/1973	Bentley	5,513,865	A	5/1996	Brooks et al.
4,003,582	A	1/1977	Maurer	5,547,204	A	8/1996	Gamzo
4,037,852	A	7/1977	Bayer et al.	5,662,187	A	9/1997	McGovern
4,043,566	A	8/1977	Johnson	5,730,241	A	3/1998	Shyr et al.
				5,741,018	A	4/1998	Huang
				D395,479	S	6/1998	Gamzo
				5,775,452	A *	7/1998	Patmont B62K 3/002
							180/181
				5,797,466	A	8/1998	Gendle
				5,829,543	A	11/1998	Diaz
				5,848,660	A	12/1998	McGreen
				5,860,657	A	1/1999	Kroher
				D410,268	S	5/1999	Yung
				5,915,707	A	6/1999	Steffen
				5,950,754	A	9/1999	Ondrish, Jr.
				5,971,411	A	10/1999	Jones et al.
				5,997,018	A	12/1999	Lee
				D421,282	S	2/2000	Orr
				6,029,763	A	2/2000	Swisher
				6,050,357	A	4/2000	Staelin et al.
				D424,979	S	5/2000	Hanagan et al.
				6,059,062	A	5/2000	Staelin
				6,070,885	A	6/2000	Ferone
				6,079,724	A	6/2000	Lin
				6,139,035	A	10/2000	Tsai
				D433,718	S	11/2000	Mcgreen
				6,158,752	A	12/2000	Kay
				D435,873	S	1/2001	Lee
				D435,874	S	1/2001	Cheng
				6,179,307	B1	1/2001	Mao
				6,189,898	B1	2/2001	Benoit
				D438,911	S	3/2001	Chen
				D438,912	S	3/2001	Barrera
				D439,623	S	3/2001	Barrera
				6,199,652	B1	3/2001	Mastroianni et al.
				6,206,387	B1	3/2001	Tsai
				6,206,388	B1	3/2001	Ouboter
				6,227,324	B1	5/2001	Sauve
				6,234,501	B1	5/2001	Chen
				6,241,264	B1	6/2001	Page

(56)

References Cited

U.S. PATENT DOCUMENTS

D444,824	S	7/2001	Udwin et al.	7,150,461	B2	12/2006	Schnuckle et al.	
D445,145	S	7/2001	Yang	7,156,405	B1	1/2007	Ming	
D446,259	S	8/2001	Udwin et al.	7,163,210	B1	1/2007	Rehkemper et al.	
D447,187	S	8/2001	Powers	7,204,330	B1	4/2007	Lauren	
D447,188	S	8/2001	Lan	7,214,337	B2	5/2007	Grande	
6,270,095	B1	8/2001	Chang	7,226,063	B2	6/2007	Paddock	
6,273,205	B1	8/2001	Tsai	7,232,139	B2	6/2007	Cole	
6,283,485	B1	9/2001	Tsai	7,351,265	B2	4/2008	Vitale et al.	
6,286,632	B1	9/2001	Chai	D569,447	S	5/2008	Thomas	
6,286,843	B1	9/2001	Lin	D574,296	S	8/2008	Keski-Luopa	
6,296,082	B1	10/2001	Tsai	7,419,171	B1	9/2008	Ka	
6,298,952	B1	10/2001	Tsai	7,431,110	B2	10/2008	Greenwood et al.	
6,299,186	B1	10/2001	Kao et al.	D579,987	S	11/2008	Hong	
6,302,415	B1	10/2001	Wang et al.	D581,991	S	12/2008	Armand et al.	
D450,355	S	11/2001	Chan	7,584,974	B2	9/2009	Jackman et al.	
D452,284	S	12/2001	McGinnis	D603,463	S	11/2009	Jessie, Jr.	
D453,198	S	1/2002	Dudley	7,610,972	B2	11/2009	Adams et al.	
D453,804	S	2/2002	Robinson	7,654,356	B2	2/2010	Wu	
6,345,678	B1	2/2002	Chang	7,712,937	B2 *	5/2010	Grossman	B62K 3/002 362/249.02
6,345,827	B1	2/2002	Benoit	D623,701	S	9/2010	Dalgard et al.	
D454,377	S	3/2002	Hsu et al.	7,900,731	B2	3/2011	McKinzie	
6,352,270	B1	3/2002	Wu	7,976,035	B2	7/2011	Chan	
D456,460	S	4/2002	Tseng	7,980,567	B2	7/2011	Yu et al.	
6,378,880	B1	4/2002	Lin	7,988,159	B2	8/2011	Chen	
6,382,366	B1	5/2002	Chang	D645,522	S	9/2011	Powers et al.	
6,386,330	B1	5/2002	Wei	8,025,300	B1	9/2011	Jordan	
6,394,213	B1	5/2002	Tsai	D654,963	S	2/2012	Powers et al.	
6,406,039	B1	6/2002	Chen	D656,560	S	3/2012	Patterson et al.	
6,409,190	B1	6/2002	Tsai	8,146,947	B2	4/2012	Hadley	
D459,761	S	7/2002	Chen	8,186,693	B2	5/2012	Kortschot	
6,416,060	B1	7/2002	Chen	8,251,377	B2	8/2012	Green et al.	
6,416,063	B1	7/2002	Stillinger et al.	8,292,018	B2	10/2012	Huang	
6,428,021	B1	8/2002	Tung	8,292,308	B2	10/2012	Green et al.	
6,428,050	B1	8/2002	Brandley et al.	D671,600	S	11/2012	Horne	
6,431,568	B1	8/2002	McCleese	D672,400	S	12/2012	Pizzi	
6,435,528	B1	8/2002	Tsai	8,348,284	B2	1/2013	Green	
6,443,470	B1	9/2002	Ulrich et al.	D678,124	S	3/2013	Canni et al.	
D464,379	S	10/2002	Lin	D681,759	S	5/2013	Treadway et al.	
6,462,493	B2	10/2002	Lan	D684,217	S	6/2013	Hadley	
6,467,560	B1	10/2002	Anderson	D686,674	S	7/2013	Hadley et al.	
6,481,728	B2	11/2002	Chen	8,490,985	B2	7/2013	Landau	
6,481,729	B2	11/2002	Herman et al.	D687,499	S	8/2013	Horne	
6,481,913	B2	11/2002	Chen	D691,671	S	10/2013	Chen et al.	
D468,236	S	1/2003	Pollitzer et al.	D692,963	S	11/2013	Radtke et al.	
D471,597	S	3/2003	Chen	D693,414	S	11/2013	Hadley et al.	
D471,598	S	3/2003	Chen	D695,644	S	12/2013	Timonen	
6,533,054	B1	3/2003	Fey	D698,868	S	2/2014	Chan	
D472,596	S	4/2003	Schnuckle et al.	8,662,508	B2 *	3/2014	Grossman	A63C 17/26 280/87.041
D473,905	S	4/2003	Schnuckle et al.	8,668,039	B2	3/2014	Tuli	
D475,093	S	5/2003	Furter et al.	8,672,074	B2	3/2014	Ganeous	
6,557,861	B2	5/2003	Saylor	8,684,121	B2	4/2014	Treadway	
6,557,873	B2	5/2003	Nardone	8,727,359	B2	5/2014	Green	
6,604,593	B1	8/2003	Mullet	D710,949	S	8/2014	Huang	
6,612,592	B1	9/2003	Soo	8,801,009	B2 *	8/2014	Sapir	B62K 3/002 280/87.05
6,619,416	B2	9/2003	Lan	8,813,892	B2	8/2014	Hadley et al.	
6,619,679	B2	9/2003	Lan	D714,396	S	9/2014	Talios	
D482,750	S	11/2003	Magerer et al.	D715,870	S	10/2014	Talios	
6,672,607	B2	1/2004	Chung	8,851,215	B2	10/2014	Goto	
6,676,138	B1	1/2004	Rosso	8,870,200	B2	10/2014	Chen	
6,679,505	B2	1/2004	Yang	D717,881	S	11/2014	Radtke et al.	
D486,532	S	2/2004	Christianson	8,925,935	B2	1/2015	Powers et al.	
6,739,421	B1	5/2004	Miya	8,985,602	B2	3/2015	Chan	
D492,367	S	6/2004	Dennis	D727,833	S	4/2015	Talios et al.	
6,796,394	B1	9/2004	Lin	9,027,690	B2	5/2015	Chavand	
D497,397	S	10/2004	Sramek	9,045,189	B2	6/2015	Lovley, II et al.	
6,832,660	B2	12/2004	Dodd	9,056,241	B2	6/2015	Green	
6,863,283	B1	3/2005	Houston et al.	D736,324	S	8/2015	Lu et al.	
D505,469	S	5/2005	Joung et al.	9,114,821	B1	8/2015	Xu	
D513,629	S	1/2006	Sramek	D738,435	S	9/2015	O'Connell	
6,981,711	B2	1/2006	Seta	D741,969	S	10/2015	Green	
D516,132	S	2/2006	Sramek	D748,738	S	2/2016	Hadley et al.	
7,007,977	B1	3/2006	Gallagher	D750,180	S	2/2016	Edlauer	
7,040,443	B1 *	5/2006	Roth	9,321,504	B2	4/2016	Hadley et al.	
				D755,899	S	5/2016	Castrucci et al.	
				D756,464	S	5/2016	Edlauer	
D527,536	S	9/2006	Mcoy	D756,465	S	5/2016	Byrne et al.	

(56)

References Cited

U.S. PATENT DOCUMENTS

9,339,720 B2	5/2016	Kortschot	D957,532 S	7/2022	Wang et al.
D764,615 S	8/2016	Treadway et al.	D958,257 S	7/2022	Cai et al.
9,452,345 B2	9/2016	Doerksen et al.	D964,473 S	9/2022	Lu
D772,990 S	11/2016	Allais	D972,040 S	12/2022	Lisle et al.
D774,145 S	12/2016	Hadley et al.	D976,333 S	1/2023	Chen et al.
D774,601 S	12/2016	Mizrahi et al.	D977,018 S	1/2023	Lu
D777,844 S	1/2017	Connell	11,565,169 B2	1/2023	Kortschot
D778,806 S	2/2017	Welford	D977,602 S	2/2023	Desberg et al.
9,592,876 B2	3/2017	Lovley, II et al.	11,697,469 B2	7/2023	Desberg et al.
9,616,318 B2	4/2017	Rogers	11,697,470 B1	7/2023	Wang
9,630,085 B2	4/2017	Treadway et al.	11,707,666 B2	7/2023	Zhang et al.
9,660,500 B2	5/2017	Huang	D995,651 S	8/2023	Hadley
9,682,309 B2	6/2017	Huang	D995,652 S	8/2023	Huang
9,764,218 B2	9/2017	Treadway et al.	D1,010,013 S	1/2024	Hadley
9,808,705 B2	11/2017	Treadway et al.	D1,020,912 S	4/2024	Desberg et al.
D806,176 S	12/2017	Peng	11,975,794 B2	5/2024	Huang
D810,623 S	2/2018	Adcock	D1,029,949 S	6/2024	Huang et al.
D810,836 S	2/2018	Hadley	12,011,654 B2	6/2024	Desberg et al.
D811,489 S	2/2018	Hadley et al.	D1,039,060 S	8/2024	Hadley
9,919,200 B2	3/2018	Mo	12,054,221 B2	8/2024	Hadley et al.
D815,215 S	4/2018	Desberg	12,059,971 B2	8/2024	Desberg et al.
9,950,244 B1	4/2018	Sargis	D1,050,269 S	11/2024	McDonald
D817,228 S	5/2018	Kroll et al.	12,145,686 B2	11/2024	Desberg et al.
D818,541 S	5/2018	Huang	D1,053,956 S	12/2024	McDonald
9,956,473 B2	5/2018	Chiu	D1,066,512 S	3/2025	Hadley
9,987,547 B2	6/2018	Ruschkowski	12,257,492 B2	3/2025	Kortschot
D824,805 S	8/2018	Karka	D1,072,062 S	4/2025	Alferez et al.
10,071,303 B2	9/2018	Pikulski	2001/0030405 A1	10/2001	Wu et al.
D829,826 S	10/2018	Liu et al.	2002/0000339 A1*	1/2002	Tsai B62J 43/28 180/220
D830,468 S	10/2018	Liu	2002/0005309 A1	1/2002	Patmont et al.
10,099,745 B2	10/2018	Hadley et al.	2002/0066610 A1	6/2002	Tsai
D834,097 S	11/2018	Lin et al.	2002/0067014 A1	6/2002	Heringer et al.
10,124,851 B2	11/2018	Lovley, II et al.	2002/0074756 A1	6/2002	Tsai
10,137,356 B2	11/2018	Kortschot	2002/0079662 A1	6/2002	Lee
D836,727 S	12/2018	Wilson et al.	2002/0079855 A1	6/2002	Parks
10,144,480 B2	12/2018	Tan et al.	2002/0093161 A1	7/2002	Udwin et al.
D837,303 S	1/2019	Desberg	2002/0096843 A1	7/2002	Wei
D839,358 S	1/2019	Cao	2002/0096850 A1	7/2002	Lu
D839,359 S	1/2019	Hadley	2002/0105152 A1	8/2002	Miller
10,189,533 B2	1/2019	Lovley, II et al.	2002/0108798 A1	8/2002	Huntsberger et al.
D842,409 S	3/2019	Novick	2002/0109313 A1	8/2002	Johnsen
D842,758 S	3/2019	Hall et al.	2002/0117825 A1	8/2002	Ho
D849,153 S	5/2019	Zhang et al.	2002/0134604 A1	9/2002	Lan
D849,601 S	5/2019	Li et al.	2002/0140194 A1	10/2002	Shaw
D855,115 S	7/2019	Yan	2002/0145264 A1	10/2002	Hung
D858,646 S	9/2019	Yang	2002/0149165 A1	10/2002	Lin
D861,538 S	10/2019	Frahm	2002/0158437 A1	10/2002	Carbonero
D866,675 S	11/2019	Desberg	2002/0167146 A1	11/2002	Chang
D872,192 S	1/2020	Hadley	2002/0170763 A1	11/2002	Townsend
D873,349 S	1/2020	Huang	2002/0180169 A1	12/2002	Kwok
10,737,743 B2	8/2020	Hadley et al.	2003/0020244 A1	1/2003	Sung
D901,595 S	11/2020	Chen	2003/0029652 A1*	2/2003	Lan B62M 6/90 180/68.5
D905,794 S	12/2020	Liu et al.	2003/0042058 A1	3/2003	Chen
D910,121 S	2/2021	Liu et al.	2003/0127816 A1	7/2003	Schnuckle et al.
D910,766 S	2/2021	Hadley	2003/0168273 A1	9/2003	Ducharme et al.
D911,455 S	2/2021	Huang	2003/0221888 A1	12/2003	McKinney, Jr. et al.
D912,180 S	3/2021	Desberg et al.	2004/0041359 A1	3/2004	Im
D914,102 S	3/2021	Hadley et al.	2004/0050603 A1	3/2004	Jaeger
10,933,299 B2	3/2021	Zhang et al.	2004/0129472 A1	7/2004	Cheng
D916,985 S	4/2021	Wu et al.	2004/0217562 A1	11/2004	Haugen et al.
D921,765 S	6/2021	Ostergaard	2005/0006859 A1	1/2005	Farrelly et al.
D923,715 S	6/2021	Wang et al.	2005/0006866 A1	1/2005	Chen
11,033,799 B2	6/2021	Kortschot	2005/0012289 A1	1/2005	Wang et al.
11,038,235 B2	6/2021	Ye	2005/0012290 A1	1/2005	McClain
11,084,550 B2	8/2021	Aguilar Ruelas et al.	2005/0116430 A1	6/2005	Chen
D936,151 S	11/2021	Lisle et al.	2005/0121866 A1	6/2005	Kamen et al.
D936,152 S	11/2021	Yao et al.	2005/0127629 A1	6/2005	Nelson et al.
D936,750 S	11/2021	Wu et al.	2005/0173175 A1	8/2005	Lee
D940,251 S	1/2022	Wang et al.	2005/0194759 A1	9/2005	Chen
D941,928 S	1/2022	Hadley	2005/0229698 A1	10/2005	Beecroft et al.
D941,929 S	1/2022	Huang	2006/0049595 A1	3/2006	Crigler et al.
D941,930 S	1/2022	Hadley	2006/0125200 A1	6/2006	Cole
D949,748 S	4/2022	Lu	2006/0279055 A1	12/2006	Morabito
11,305,830 B2	4/2022	Bartolotta et al.	2007/0045976 A1	3/2007	Wu
11,338,883 B2	5/2022	Hadley et al.	2007/0126191 A1	6/2007	Kay, III
			2008/0197594 A1	8/2008	Ling
			2008/0203691 A1	8/2008	Hsu

(56)

References Cited**U.S. PATENT DOCUMENTS**

2009/0071740	A1	3/2009	Palmer	
2009/0160150	A1	6/2009	Johnson	
2009/0200768	A1	8/2009	Armand et al.	
2009/0273152	A1	11/2009	Chung	
2009/0321193	A1	12/2009	Jackman et al.	
2010/0090422	A1	4/2010	Chan et al.	
2010/0123295	A1	5/2010	Landau	
2010/0133765	A1	6/2010	Chen	
2011/0011657	A1	1/2011	Donnell et al.	
2011/0031711	A1	2/2011	Grossman	
2011/0056177	A1	3/2011	Goto	
2012/0086181	A1	4/2012	Kinnaman	
2013/0026734	A1	1/2013	Zhang et al.	
2013/0081891	A1	4/2013	Ulmen et al.	
2013/0087983	A1	4/2013	Ngai	
2013/0186698	A1	7/2013	Sarokhan	
2013/0270016	A1	10/2013	Donnell et al.	
2013/0282216	A1	10/2013	Edney	
2013/0320648	A1	12/2013	Eckert et al.	
2014/0090916	A1	4/2014	Lovley, II et al.	
2014/0158446	A1	6/2014	Treadway	
2014/0196968	A1*	7/2014	Bieler	B62M 6/65 180/181
2014/0332297	A1	11/2014	Kortschot	
2015/0196831	A1	7/2015	Treadway et al.	
2015/0306511	A1	10/2015	Norman et al.	
2015/0352430	A1	12/2015	Treadway et al.	
2016/0129963	A1	5/2016	Ying et al.	
2016/0213999	A1	7/2016	Treadway et al.	
2016/0256767	A1	9/2016	Cerboneschi	
2016/0311322	A1	10/2016	Lagant et al.	
2016/0318578	A1	11/2016	Heber et al.	
2017/0182397	A1	6/2017	Zhang	
2017/0182398	A1	6/2017	Garcia Elena	
2017/0240239	A1	8/2017	Huang et al.	
2017/0259162	A1	9/2017	Mo	
2018/0029659	A1	2/2018	Desberg	
2019/0084605	A1	3/2019	Berendzen et al.	
2019/0263468	A1*	8/2019	Huang	B62K 15/00
2020/0307735	A1	10/2020	Paris et al.	
2021/0236910	A1	8/2021	Behrmann	
2022/0006959	A1	1/2022	Chen et al.	
2022/0041243	A1	2/2022	Qiao	
2022/0048402	A1	2/2022	Desberg et al.	
2022/0204112	A1	6/2022	Sun et al.	
2022/0348284	A1	11/2022	Hadley et al.	
2023/0127757	A1	4/2023	Lovley, II et al.	
2023/0415024	A1	12/2023	Kortschot	
2024/0010291	A1	1/2024	Desberg et al.	
2024/0399920	A1	12/2024	Desberg et al.	
2025/0010168	A1	1/2025	Desberg et al.	

FOREIGN PATENT DOCUMENTS

CN	2273210	Y	1/1998
CN	2436196	Y	6/2001
CN	2448414	Y	9/2001
CN	2449767	Y	9/2001
CN	2452873	Y	10/2001
CN	1143702	C	3/2004
CN	2619503	Y	6/2004
CN	2717494	Y	8/2005
CN	2785659	Y	6/2006
CN	2915189	Y	6/2007
CN	200966900	Y	10/2007
CN	203486089	U	3/2014
CN	204021174	U	12/2014
CN	104787183		7/2015
CN	104941171	A	9/2015
CN	105148496	A	12/2015
CN	303684781		5/2016
CN	106422285	A	2/2017
CN	205952205		2/2017
CN	205952219	U	2/2017

CN	206138715	U	5/2017
CN	108295457	A	7/2018
CN	108325204	A	7/2018
CN	304753393		7/2018
CN	109927829		6/2019
CN	305407387		10/2019
CN	110979533	A	4/2020
CN	210258685	U	4/2020
CN	210618344	U	5/2020
CN	305886136		6/2020
CN	111703528	A	9/2020
DE	4424297	A1	1/1996
DE	201 07 388	U1	7/2001
EP	0 995 468		4/2000
EP	3943373	A1	1/2022
GB	2 441 394		9/2008
GB	9007711890-0001		2/2020
GB	8210638000-1000		4/2020
JP	56085381		7/1981
JP	H06-305476	A	11/1994
JP	2001-225775	A	8/2001
JP	3081197	U	10/2001
KR	200378761	Y1	3/2005
SU	912584	A1	3/1982
TW	M582478	U	8/2019
WO	WO 96/20762		7/1996
WO	WO 2000/03773		1/2000
WO	WO 2002/089933		11/2002
WO	WO 2008/006275		1/2008
WO	WO 2008/019482		2/2008
WO	WO 2015/095368		6/2015
WO	WO 2015/114614	A2	8/2015
WO	WO 2016/128488		8/2016
WO	WO 2017/050247		3/2017
WO	WO 2018/013994		1/2018
WO	WO 2018/019156	A1	2/2018
WO	WO 2019/0232248		12/2019
WO	WO 2020/069299	A1	4/2020
WO	WO 2021/073422	A1	4/2021
WO	WO 2022/032136		2/2022

OTHER PUBLICATIONS

Amy Tucker, "We tried both Calgary e-scooter brands Lime and Bird—here's how they compare," Jul. 30, 2019, the star.com, site visited Jan. 23, 2023, <https://www.thestar.com/calgary/2019/07/29/we-tried-both-calgary-e-scooter-brands-lime-and-bird-heres-how-they-compare.html>, in 3 pages.

Kids Roller Skates—Heel Wheel Roller Skates Attachable Shoe, <https://www.amazon.ca/Kids-Roller-Skates-Attachable-Trainer/dp/B08R21VVDJ/>, Dec. 19, 2020, in 1 page.

"Razor Electric Hub Motor Scooter Owner's Manual (Kobalt)," Aug. 12, 2020, data2.manualslib.com, site visited Jan. 23, 2023, <https://data2.manualslib.com/pdf/209/20836/20836/2083509-razor-kobalt.pdf?edd92ca8adaa31a5bed07f3ecd24e2e5>, in 4 pages.

Razor Jetts Heel Wheels, Purple, One Size, <https://www.amazon.ca/Razor-Jetts-Heel-wheels-purple/dp/B073XQJZSX/>, Jul. 14, 2017, in 1 page.

"Razor Worldwide," Nov. 11, 2019, facebook.com, site visited Jan. 23, 2023, <https://www.facebook.com/RazorWorldwide/photos/2558446474245901>, in 1 page.

Tim Nelson and Elizabeth Dunbar, "Electric scooter rentals debut in Twin Cities; Mpls. Eyes regulation," Jul. 10, 2018, mprnews.org, site visited Jan. 23, 2023, <https://www.mprnews.org/story/2018/07/10/electric-scooter-rentals-debut-twin-cities-hours-before-mpls-vote>, in 1 page.

"Xiaomi M365/PRO/PRO2/1S/3 any voltage hot swappable extra battery," Jun. 8, 2020, youtube.com, site visited Jan. 23, 2023, <https://www.youtube.com/watch?v=1cJfDFCtMGI>, in 1 page.

"Zag Daily," Aug. 25, 2020, twitter.com, suite visited Jan. 23, 2023, <https://mobile.twitter.com/ZagDaily/status/1298193918660694017/photo/1>, in 1 page.

Office Action in corresponding Japanese Patent Application No. 2018-538211, dated Aug. 10, 2021, in 6 pages.

(56)

References Cited

OTHER PUBLICATIONS

Office Action in corresponding Japanese Patent Application No. 2022-076172, dated Dec. 23, 2022, in 8 pages.

International Search Report in co-pending International Application No. PCT/US2017/013998, mailed May 25, 2017, in 4 pages.

International Preliminary Report on Patentability in co-pending International Application No. PCT/US2017/013998, issued Jul. 24, 2018, in 10 pages.

Extended European Search Report in corresponding European Patent Application No. 17741870.4, dated Jul. 25, 2019, in 13 pages.

* cited by examiner

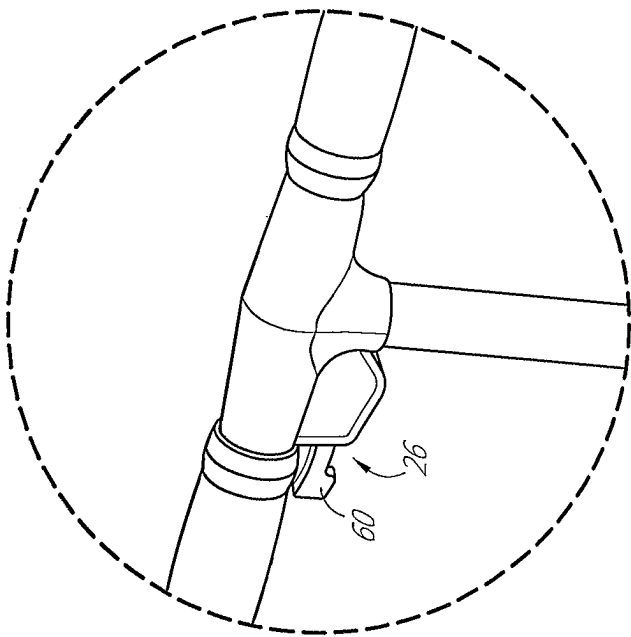


FIG. 2

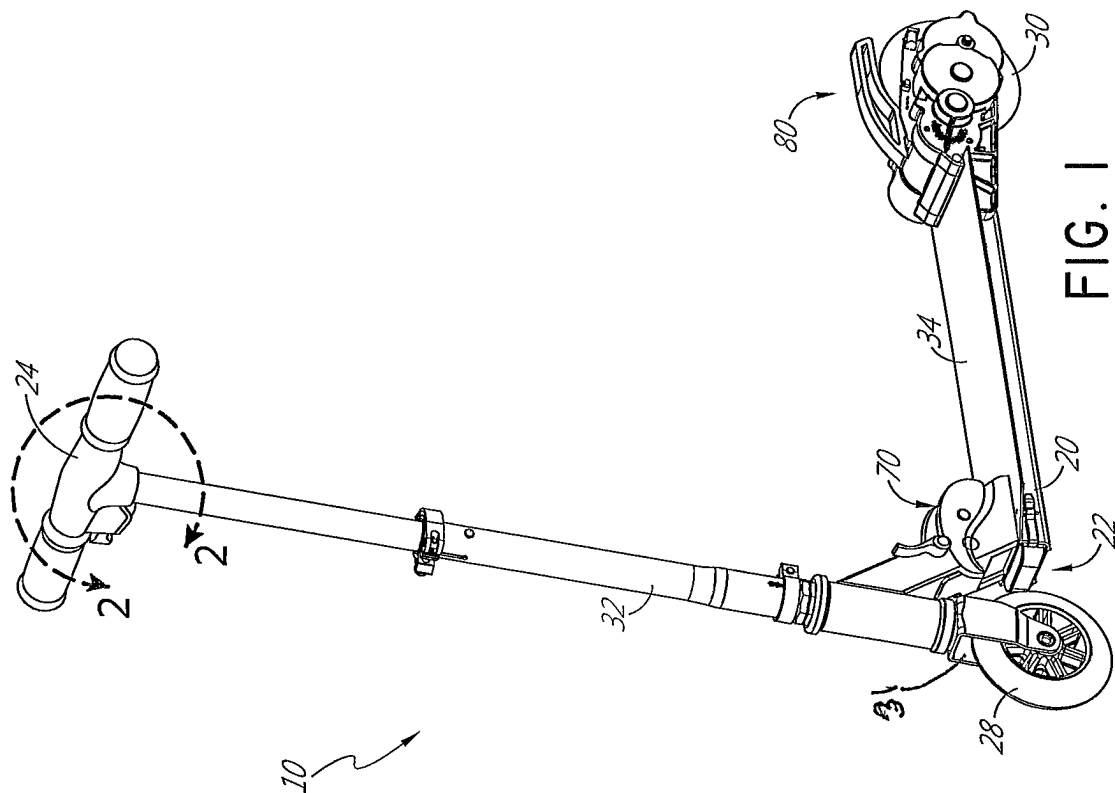
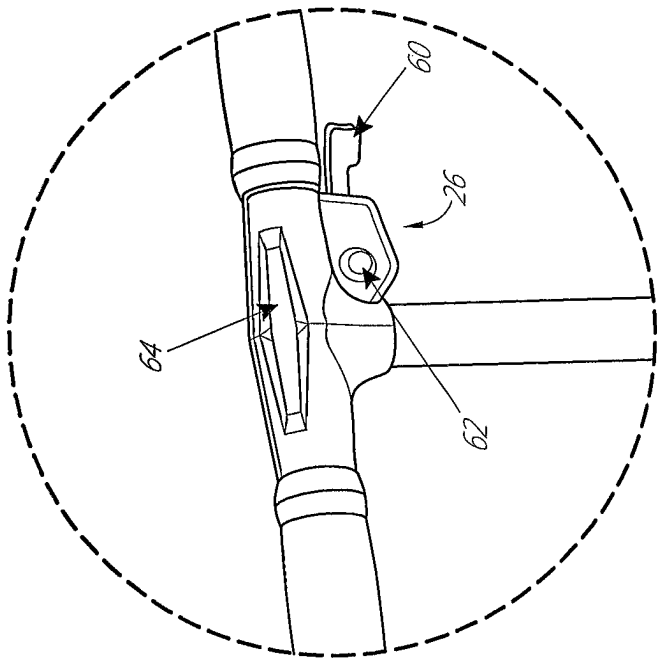
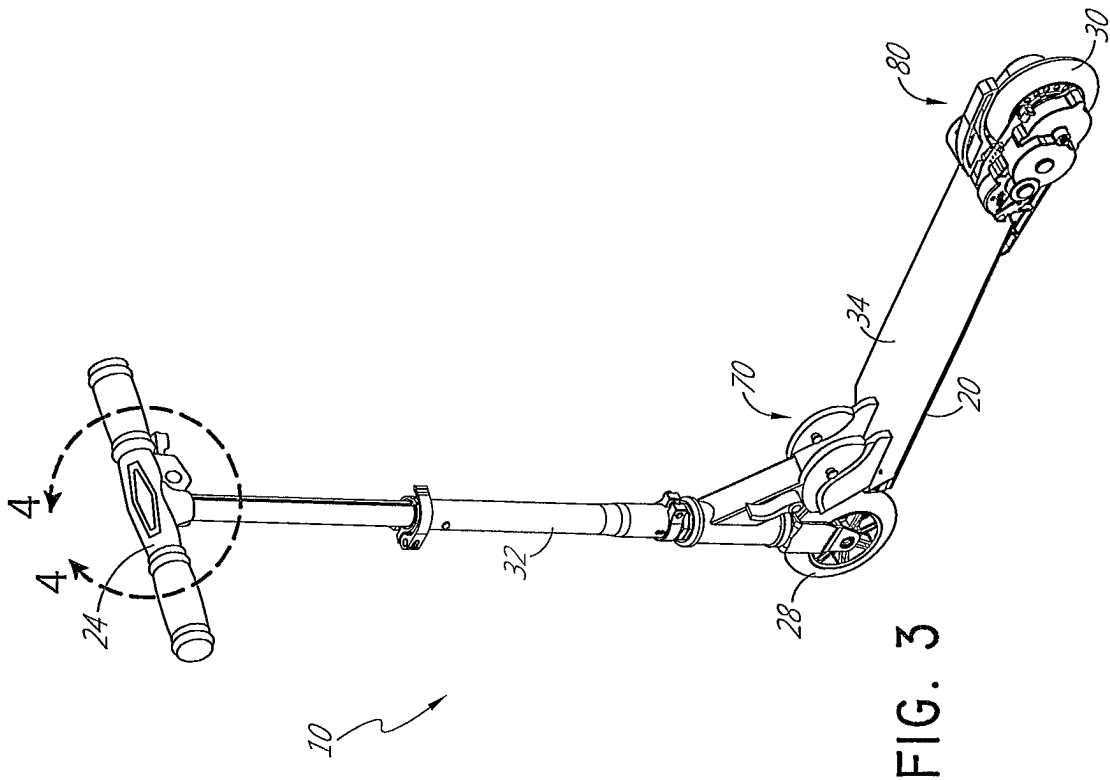


FIG. 1



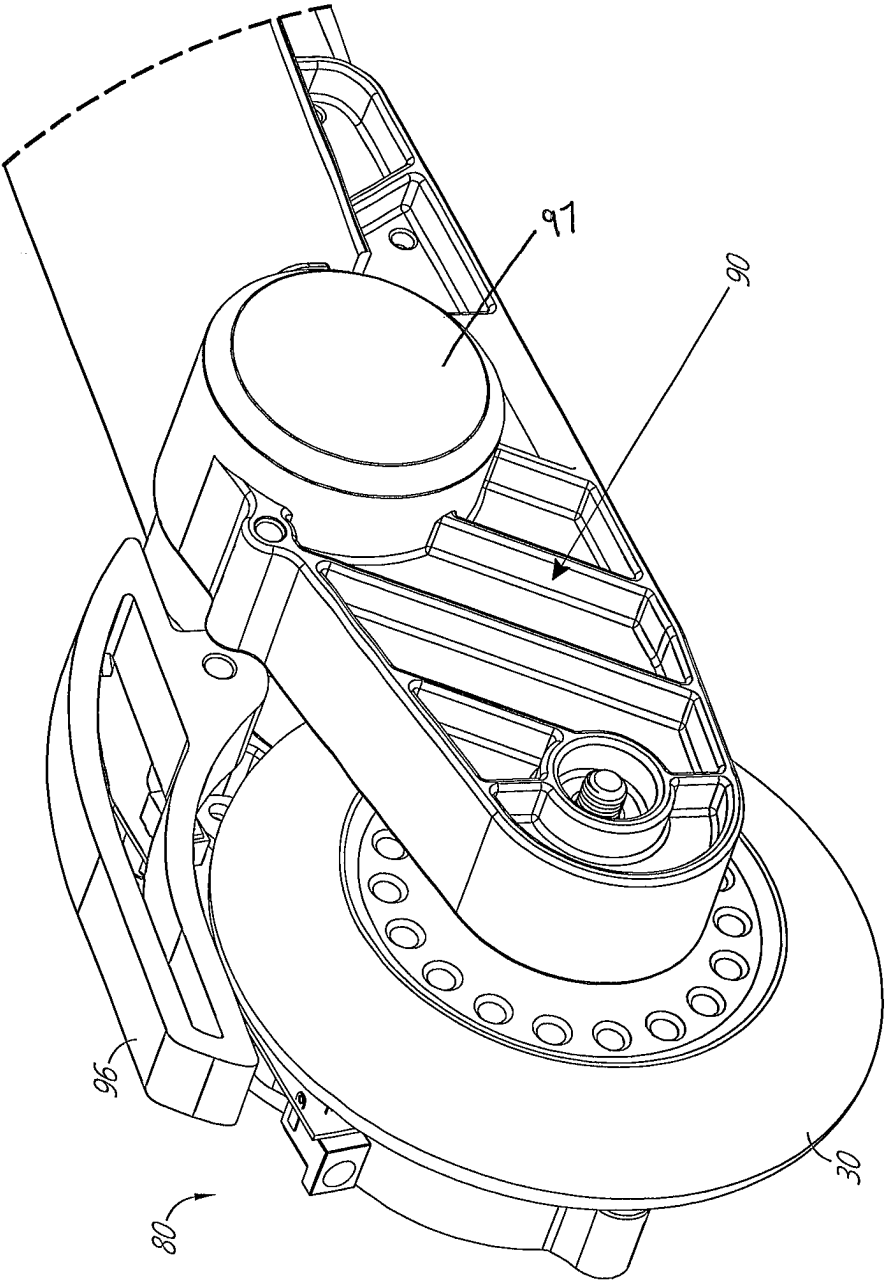


FIG. 5

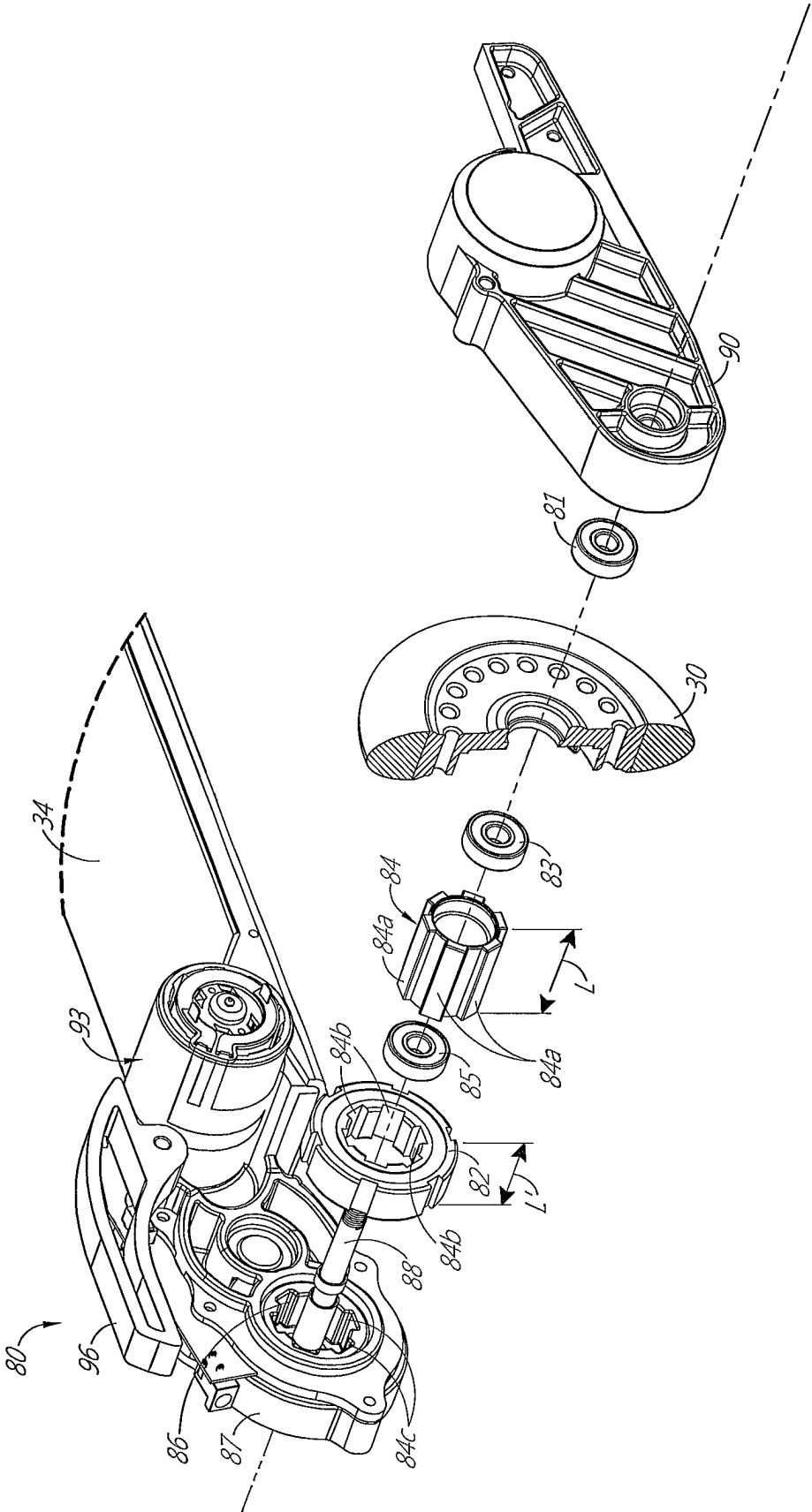
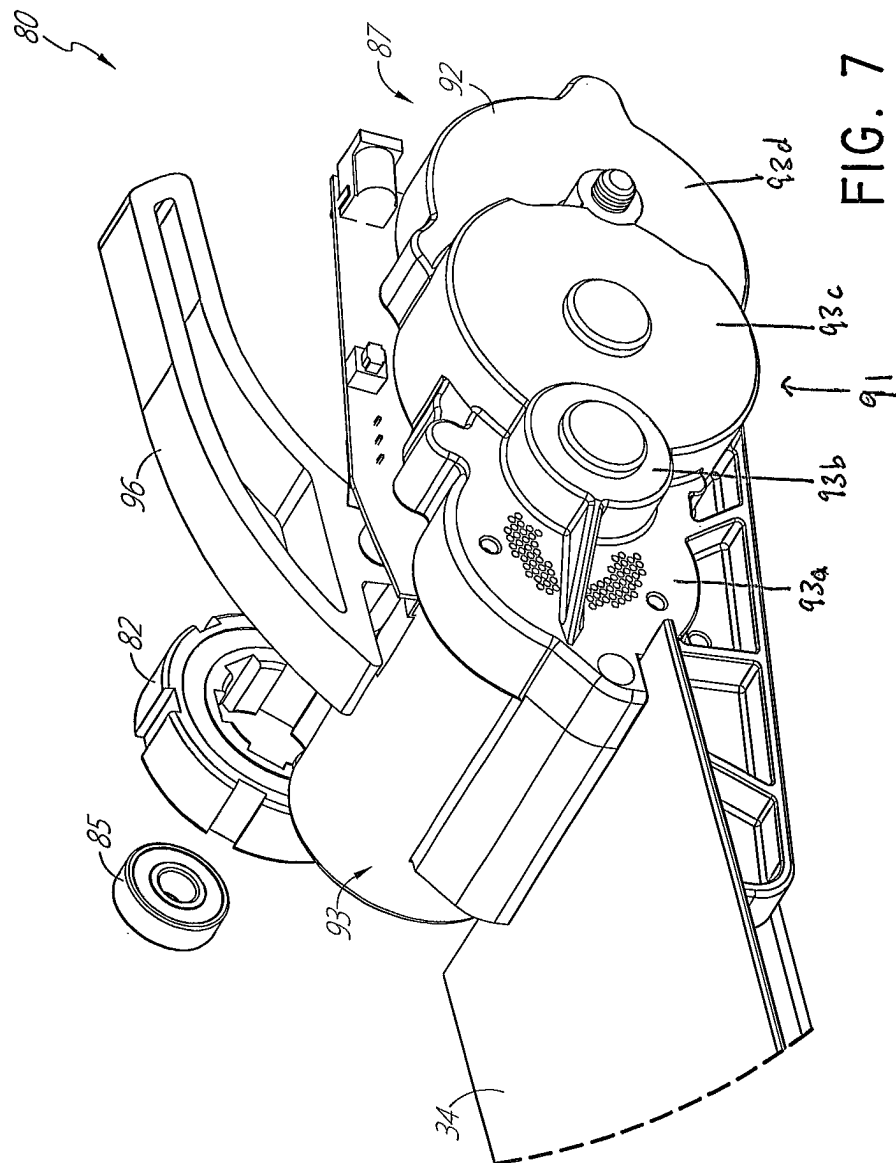


FIG. 6



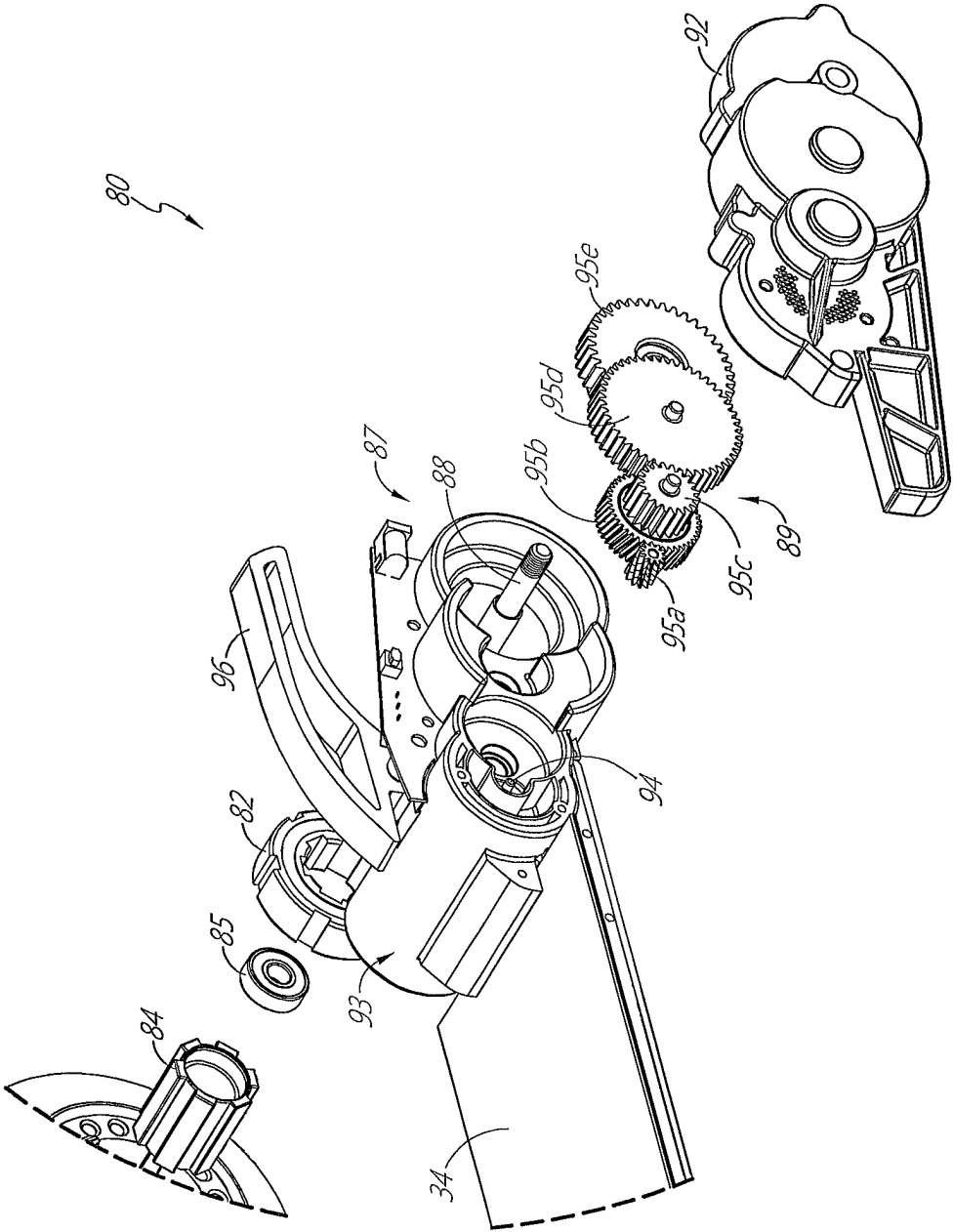


FIG. 8

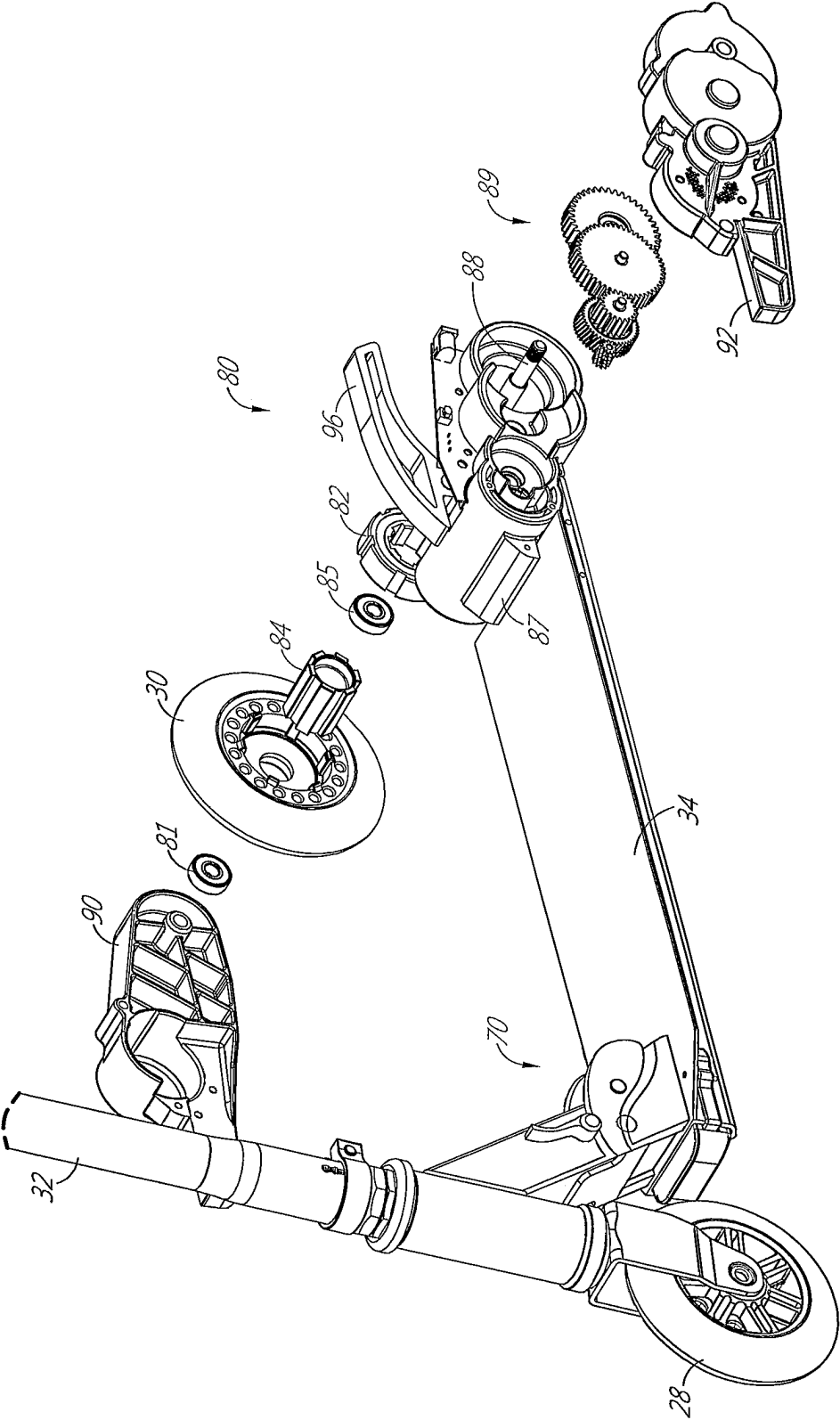


FIG. 9

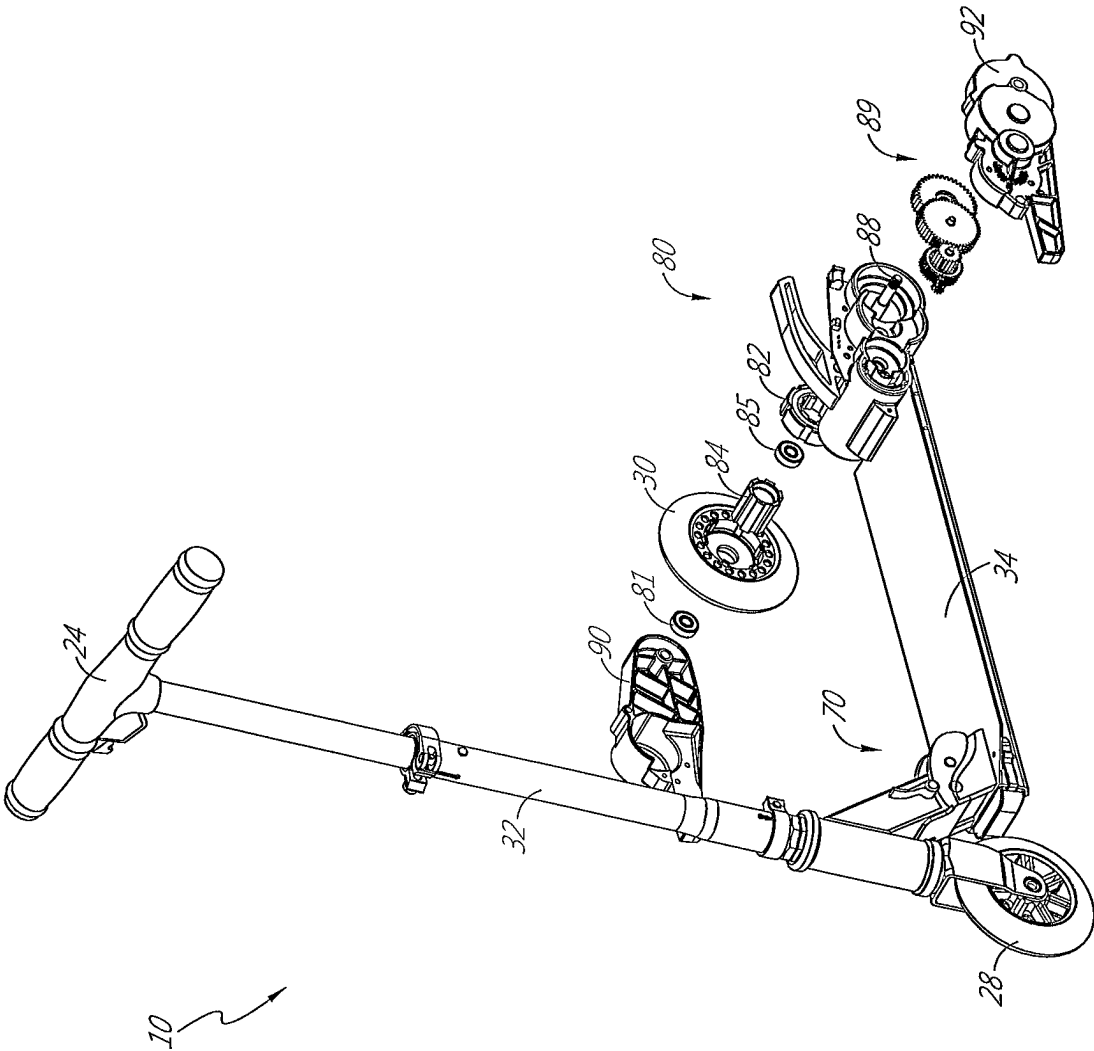


FIG. 10

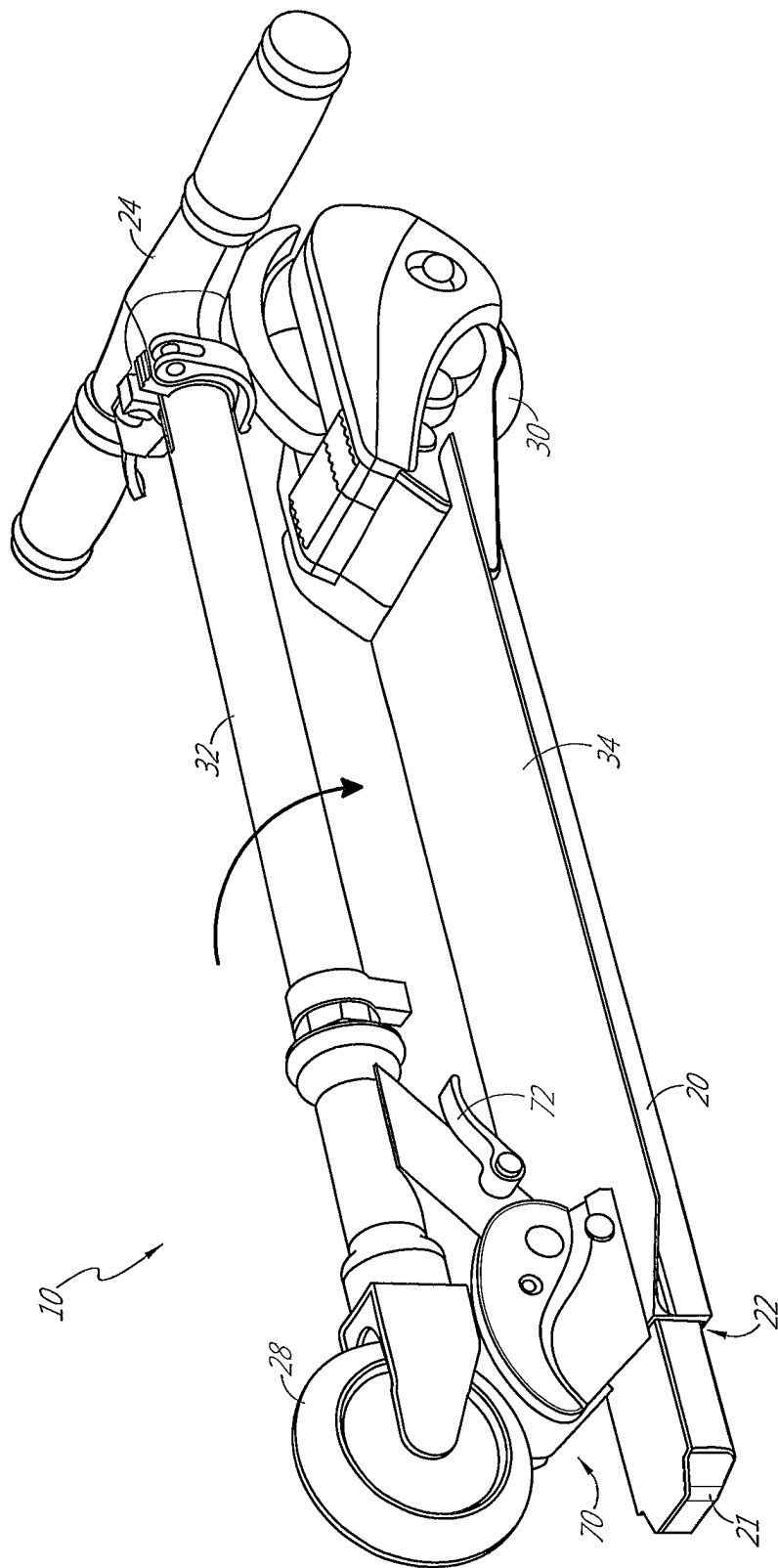


FIG. II

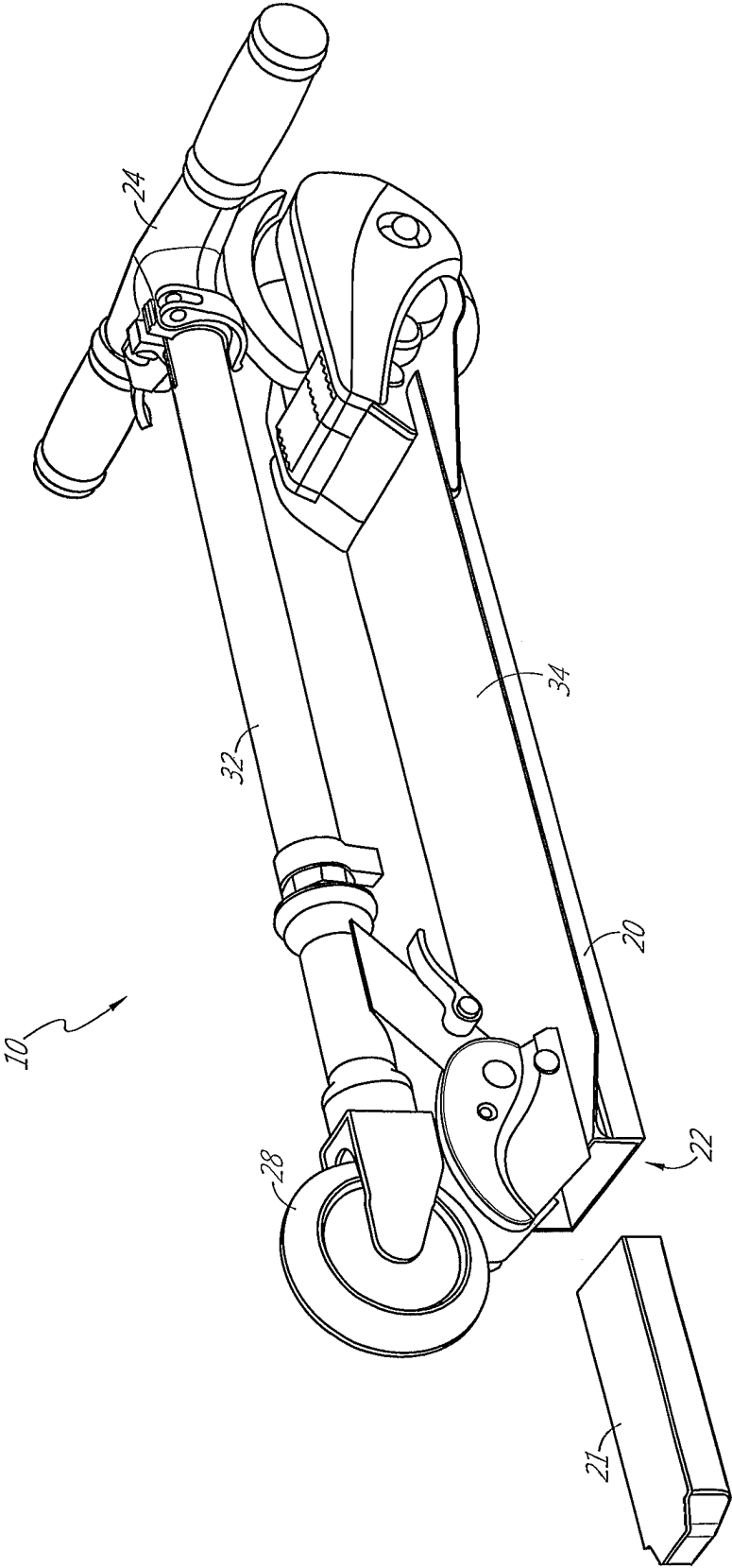


FIG. 12

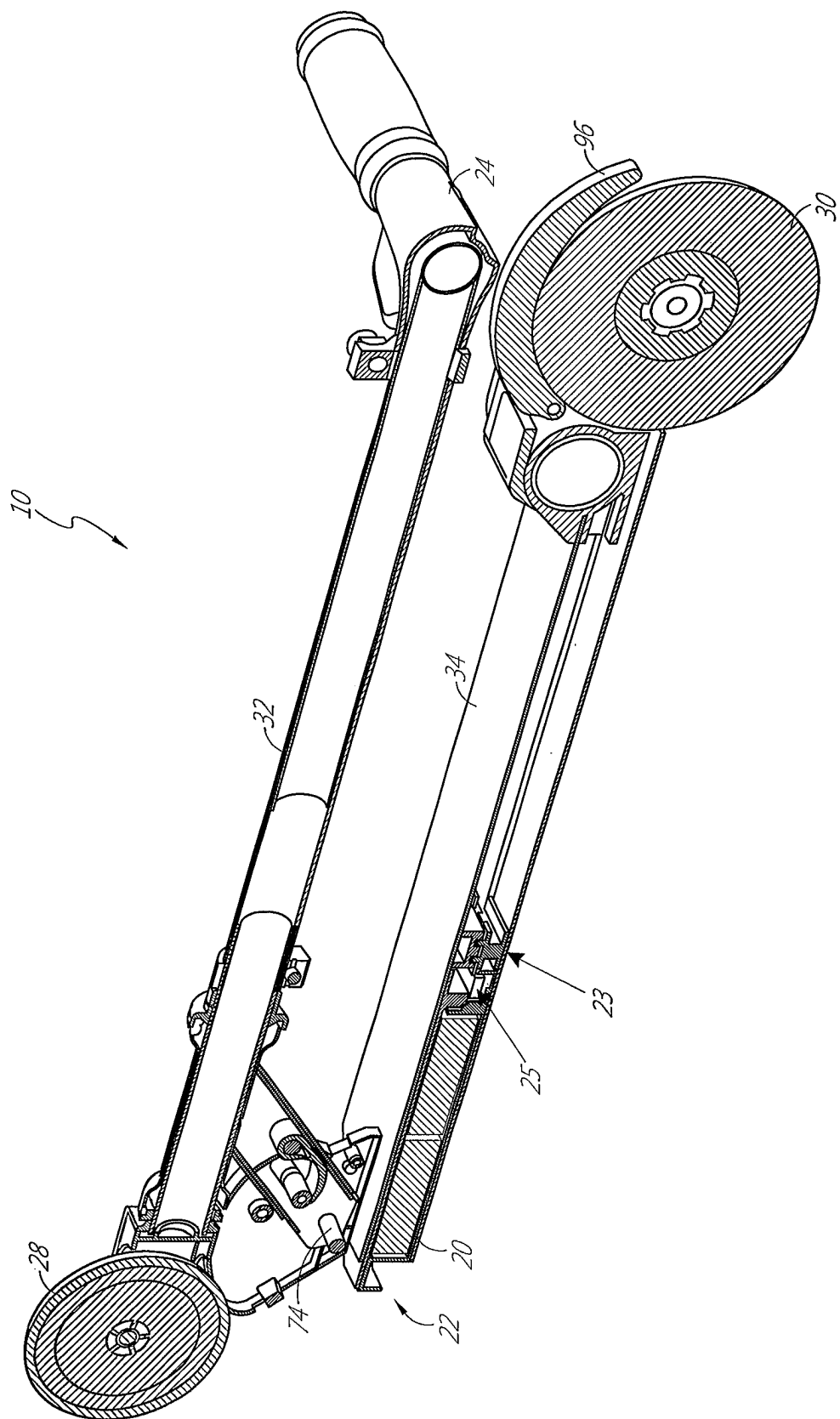


FIG. 13

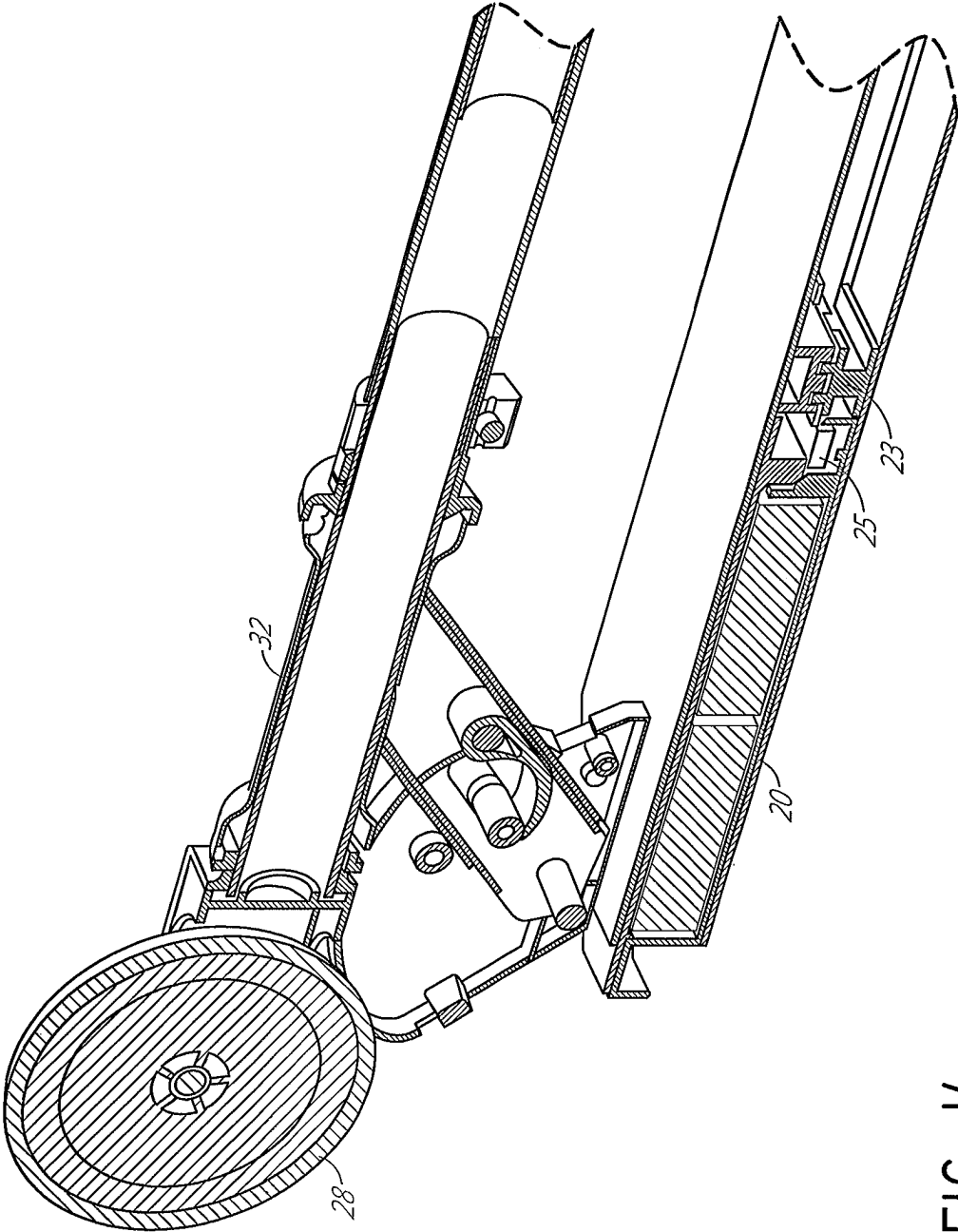
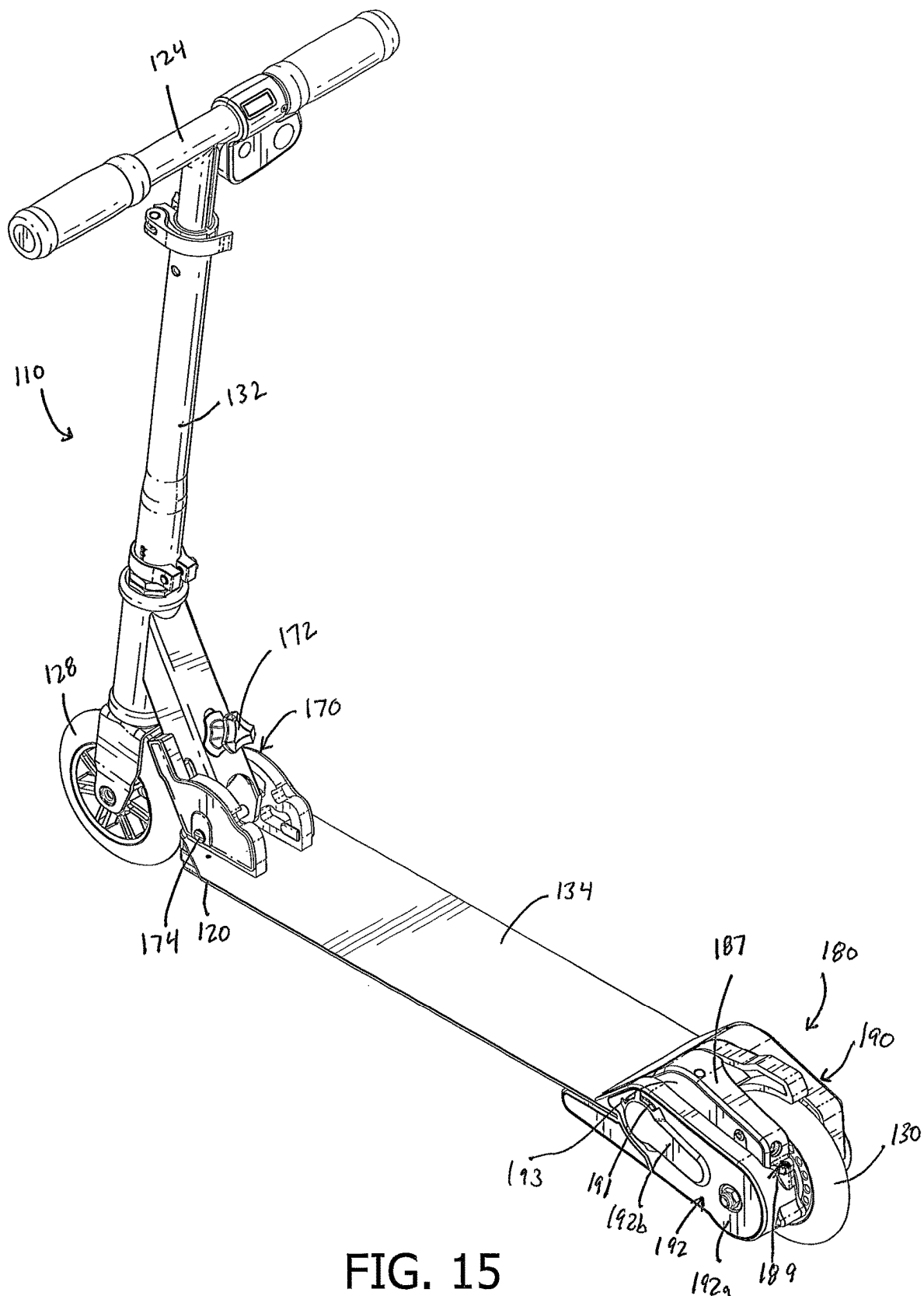


FIG. 14



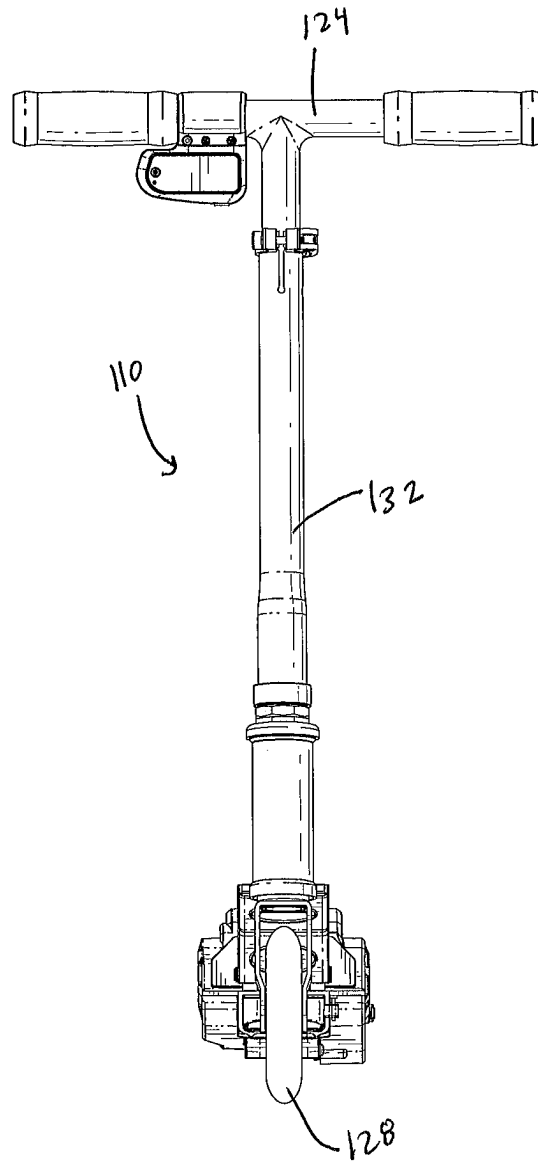


FIG. 16

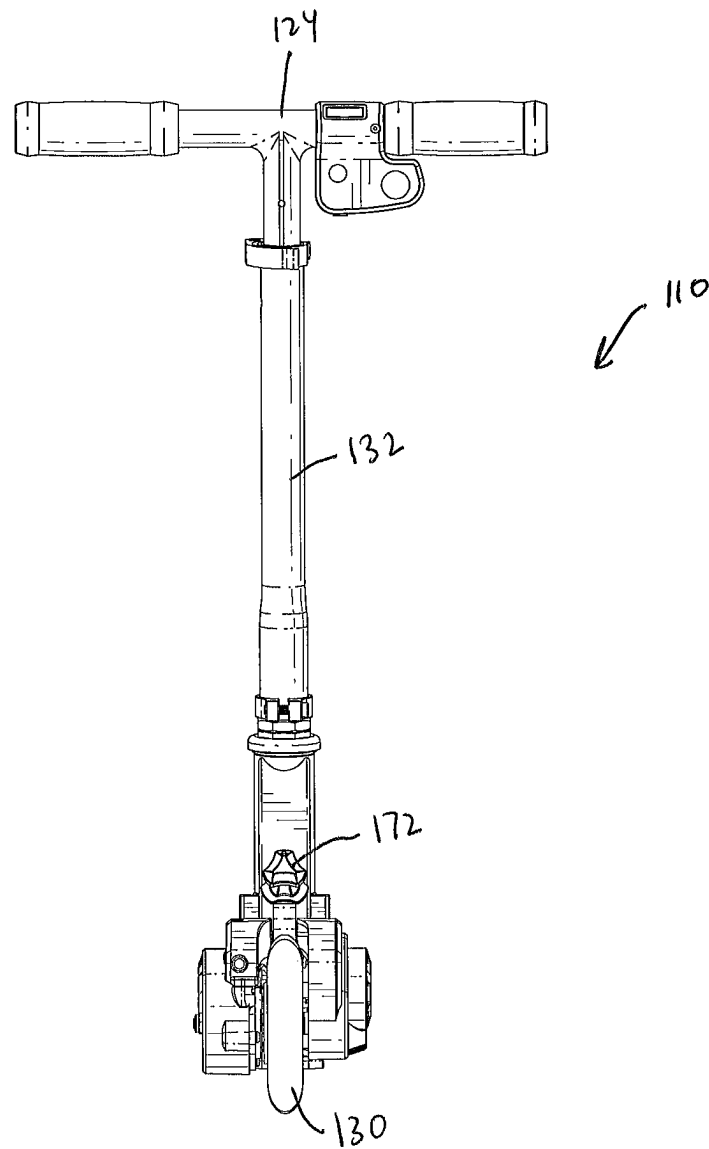


FIG. 17

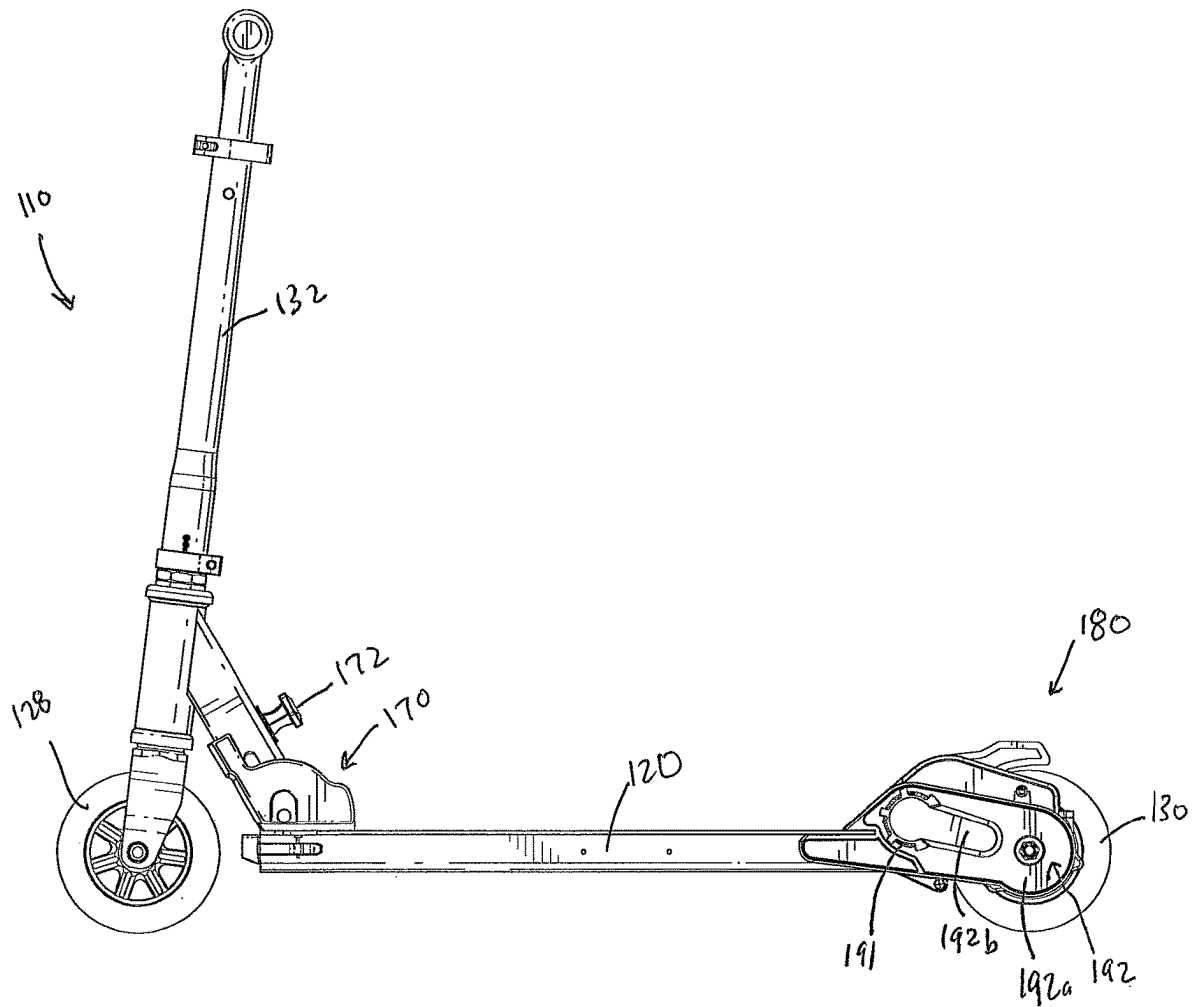


FIG. 18

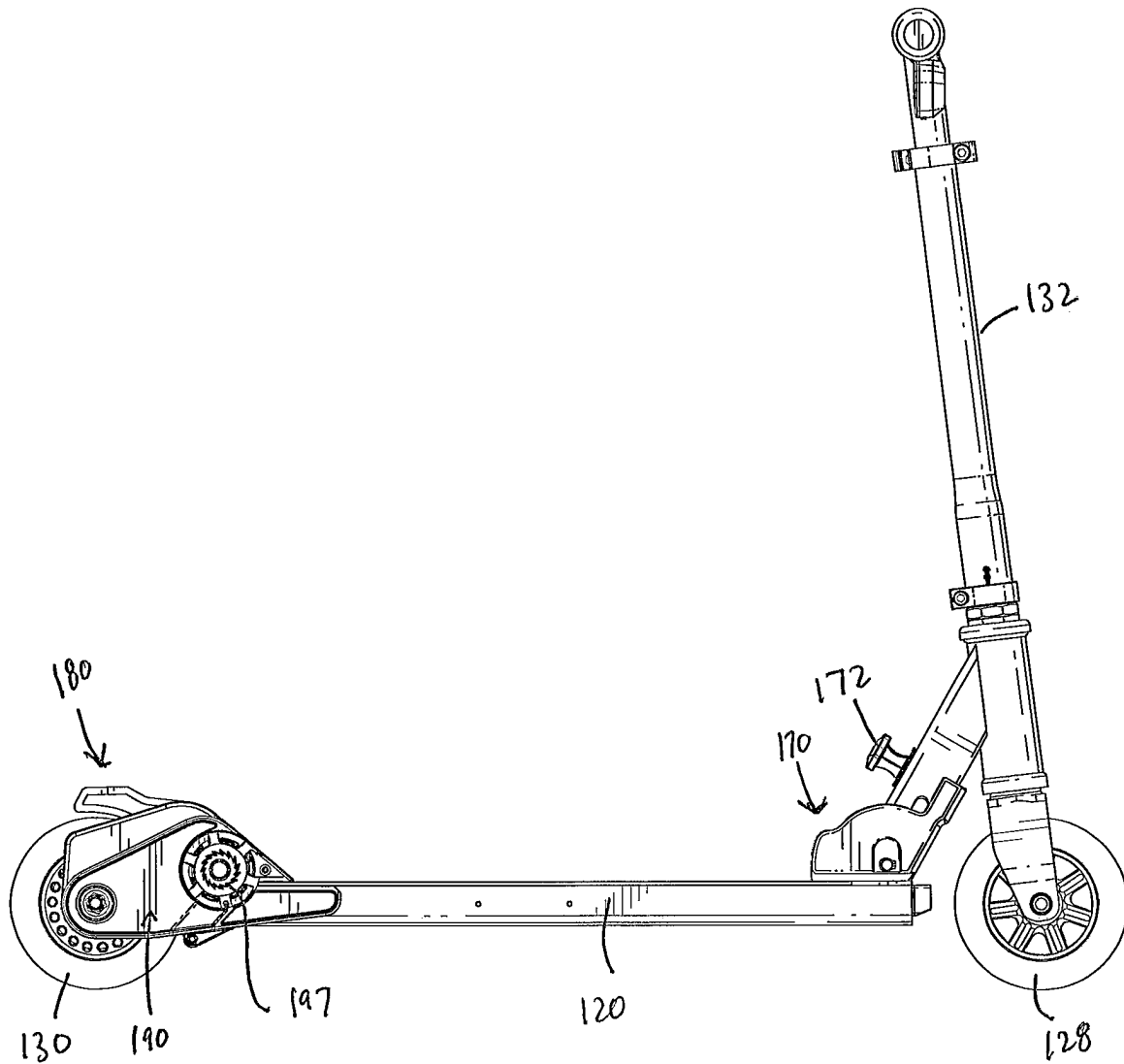
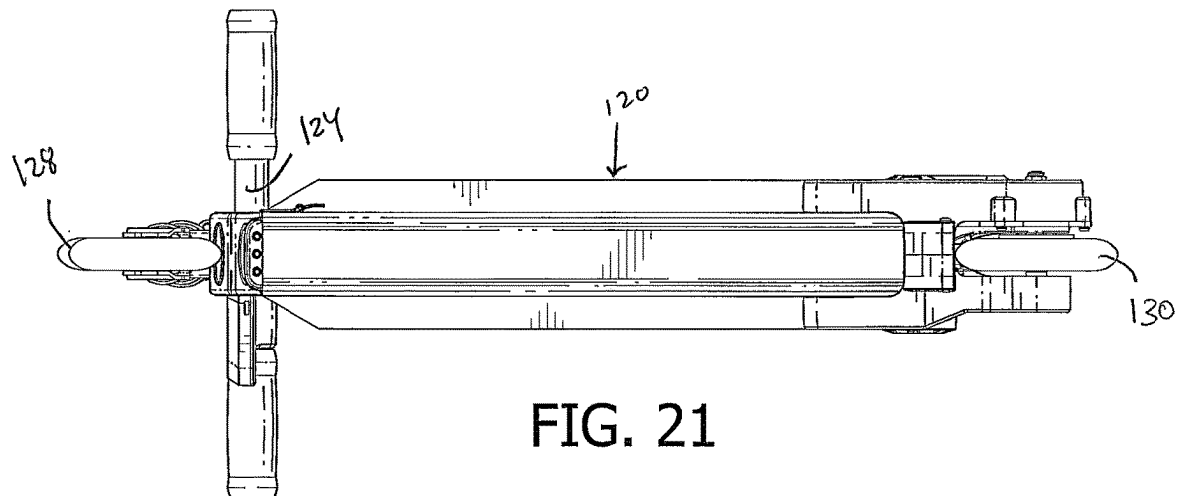
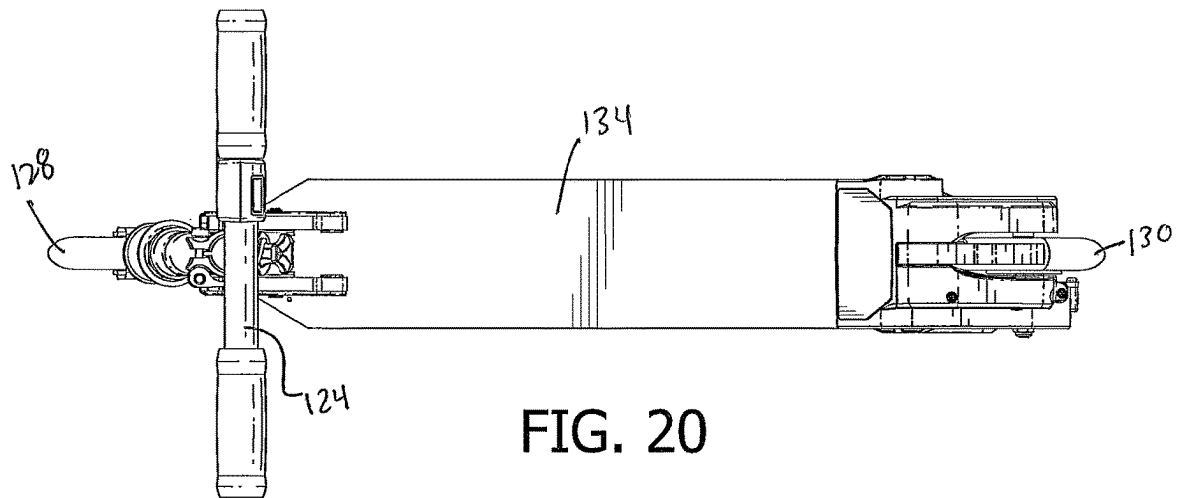


FIG. 19



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ELECTRIC SCOOTER**CROSS REFERENCE**

This application is a continuation of U.S. patent application Ser. No. 16/263,801, filed Jan. 31, 2019, now U.S. Pat. No. 11,975,794, which is a continuation of U.S. patent application Ser. No. 15/409,488, filed Jan. 18, 2017, which claims the priority benefit under at least 35 U.S.C. § 119 of U.S. Patent Application No. 62/286,124, filed Jan. 22, 2016, and U.S. Patent Application No. 62/340,978, filed May 24, 2016, and which is a continuation in part of U.S. patent application Ser. No. 29/577,038, filed Sep. 8, 2016, now Pat. No. D818,541. The entirety of each of the aforementioned applications is incorporated by reference herein.

BACKGROUND**Field**

Embodiments of the disclosure relate generally to personal mobility vehicles, such as electric scooters that can freewheel in at least one direction.

Description of Certain Related Art

The use of personal mobility vehicles, such as scooters, has become a popular recreational activity as well as useful means of personal transportation. Scooters have become popular among many age groups and there are many different scooter variations and scooter designs. Conventional scooters have at least two wheels and a steering mechanism. Scooter designs have become more compact and convenient, and some scooters have folding mechanisms allowing the user to easily store or carry the scooter. In order to allow a user to slow or stop a scooter, some scooters include a braking mechanism.

SUMMARY OF CERTAIN FEATURES

With the increased popularity of scooters, variations of the conventional scooters have gained popularity as well, including motorized scooters. With the use of electric or motorized scooters, the weight of an electric motor may limit the portability of the scooter, particularly for commuters. Additionally, use of large electric motors can require large batteries, further increasing the weight of the scooter. Thus there is a need to provide an electric scooter having a small motor and battery to allow for a lightweight and portable electric scooter at a low cost.

Some electric powered scooter embodiments include an electric motor mounted to the front or rear of the scooter frame near at least one of the front or rear wheels, with a drive assembly coupled to the motor and configured to transfer drive power from the motor to the driven front or rear wheel. The drive assembly can include powered rollers using frictional directional force upon the circumference of the driven wheel or a drive chain linking the drive means to the hub of the driven wheel. The driven wheel can be electrically driven when a throttle assembly controlling the motor, commonly located on or about the handlebar, is activated by the user. Some electric powered scooter embodiments include a battery compartment accessible when the scooter is a folded configuration.

Scooters having a driven wheel that is not able to freewheel in one direction can have several disadvantages, such as unwanted noise, undesirable friction, and/or difficulty

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moving or transporting the scooter due to the wheel being unable to move unless powered. Certain of the electric powered scooter embodiments disclosed herein address these issues, or other issues, by allowing the driven wheel to freewheel in at least one direction so that the scooter may be moved when unpowered.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, aspects, and advantages are described below with reference to drawings of an example embodiment, which are intended to illustrate, but not to limit, the present disclosure.

FIG. 1 is a perspective view of an embodiment of a scooter.

FIG. 2 is a close up view of a wireless remote throttle that may be used with the scooter of FIG. 1.

FIG. 3 is a rear perspective view of the scooter of FIG. 1.

FIG. 4 is a close up view of a display and throttle arrangement of the scooter of FIG. 1.

FIG. 5 is a close up rear perspective view of a first side of the rear wheel assembly of the scooter of FIG. 1.

FIG. 6 is a partial exploded view of the rear wheel assembly of the scooter of FIG. 1.

FIG. 7 is a close up rear perspective view of a second side of the rear wheel assembly of the scooter of FIG. 1.

FIG. 8 is a close up exploded view of the rear wheel assembly of the scooter of FIG. 1.

FIG. 9 is another exploded view of the rear wheel assembly of the scooter of FIG. 1.

FIG. 10 is another exploded view of the rear wheel assembly of the scooter of FIG. 1.

FIG. 11 is a perspective view of the scooter of FIG. 1 in a folded configuration.

FIG. 12 is a perspective view of the scooter of FIG. 11 with a battery pack removed.

FIG. 13 is a partial cross-sectional view of the scooter of FIG. 11.

FIG. 14 is a close up view of the cross-sectional view of FIG. 13.

FIG. 15 is a perspective view of another embodiment of a scooter.

FIG. 16 is a front view of the scooter of FIG. 15.

FIG. 17 is a rear view of the scooter of FIG. 15.

FIG. 18 is a left-side view of the scooter of FIG. 15.

FIG. 19 is a right-side view of the scooter of FIG. 15.

FIG. 20 is a top view of the scooter of FIG. 15.

FIG. 21 is a bottom view of the scooter of FIG. 15.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

Reference will now be made in detail to various embodiments of the present technology, which relates to various personal mobility devices, such as scooters. Although certain specific embodiments of the present technology are described, the present technology is not limited to these embodiments. On the contrary, these described embodiments are merely illustrative of the present technology, and the present technology is intended to also cover alternatives, modifications, and equivalents. Furthermore, in the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present technology. However, it will be recognized by one of ordinary skill in the art that embodiments may be practiced without these specific details. In some instances, well known methods, procedures, compounds, compositions and mecha-

nisms have not been described in detail as not to unnecessarily obscure aspects of embodiments of the present technology.

FIGS. 1-14

FIGS. 1-14 illustrate an embodiment of an electric scooter 10. As shown in FIG. 1, the scooter 10 includes a scooter body 20, a support surface 34, a rotatable shaft 32, and a handlebar assembly 24. The support surface 34 can be configured to support at least one foot of the user. The illustrated support surface 34 is configured to accommodate both feet of a user, such as in one or both of a fore-and-aft or side-by-side arrangement. The scooter 10 includes at least one front wheel 28 and at least one rear wheel 30 connected with the scooter body 20. The front wheel 28 and the rear wheel 30 can be spaced apart from one another with the support surface 34 extending therebetween. The wheels 28, 30 can be aligned in the same plane, as depicted, and located at opposite ends of the support surface 34. As depicted, some embodiments of the scooter 10 include a rotatable shaft 32 which rotates with the handlebar assembly 24 so that the handlebar assembly 24 can rotate or swivel within a head tube of the scooter body 20. The shaft 32 can be connected with a fork 31 that connects with the front wheel 28, such that the handlebar assembly 24 can be rotated to swivel the front wheel 28 to steer the scooter 10. The fork 31 can be directly connected to sides of the front wheel 28 and/or an axle that extends through a rotational axis of the front wheel 28.

In some embodiments, the scooter 10 includes a folding assembly 70 and/or a motor assembly, such as a rear drive assembly 80. The folding assembly 70 allows the handlebar assembly 24 and the rotatable shaft 32 to pivot relative to the body 20 such that the handlebar assembly 24 folds against the body 20 for compact transport and storage of the scooter 10. Accordingly, in some embodiments, the rotatable shaft 32 can pivot about the folding assembly 70.

The rear drive assembly 80 can include a motor 93 (see FIG. 6) to drive the rear wheel 30. In certain embodiments, torque from the motor 93 is transmitted to the wheel 30 via a transmission, such as a gear set or gear assembly. The folding assembly 70 and the rear drive assembly 80 are discussed in greater detail below.

As shown in FIG. 2, in some embodiments, the handlebar assembly 24 includes a control mechanism, such as a wireless throttle remote assembly 26. The wireless throttle remote assembly 26 includes a throttle 60 that can be toggled by the user to increase or decrease the speed of a motor 93 to increase or decrease the speed of the scooter. In some embodiments, the throttle assembly 26 is wireless; however, in other embodiments, the throttle assembly 26 is wired to a motor 93 and/or a battery. In some embodiments, the throttle assembly 26 may include a brake lever (not shown) to activate a braking assembly to slow or stop the scooter 10. The throttle assembly 26 can be connected to the braking assembly through a wireless and/or mechanical connection. In some embodiments, the throttle 60 comprises a button, switch, lever, or other actuator available to the hand of the user while the user operates the scooter 10. Depressing the button throttle 60 initiates a signal (e.g., an electric or RF signal) to a controller, which signals for power to be transferred from the battery to the electric motor 93, resulting in rotational power being transferred from the electric motor 93 to the at least one driven wheel. In certain

embodiments, the user is able to engage the throttle by a hand lever mechanism, or other similar hand actuated mechanisms.

With reference to FIGS. 3 and 4, in some embodiments, the throttle assembly 26 includes an on/off switch or button 62 and/or a display 64. The on/off switch 62 may be used to send a “power on” or “power off” signal to the battery and/or the motor 93. The display 64 provides a visual indication of a characteristic of the scooter, such as battery level, speed, distance, direction, or other information. In some embodiments, the display 64 indicates the amount of battery power remaining. The display 64 may indicate the battery power as a digital percentage display, or as a color gradient (such as green, yellow, red), or with any other visual indicator. The display 64 may be configured to indicate when the battery should be replaced and/or recharged, or the amount of time remaining until the battery is fully discharged.

FIG. 5 illustrates a close-up view of the rear wheel 30 and rear drive assembly 80 of the scooter 10. A side piece 90 is removable to provide access to the rear wheel 30 in case the rear wheel should need repair or replacement, such as due to wear, etc. A side piece 90 can be connected to the rear drive assembly 80 to surround at least a portion of the rear drive assembly 80. For example, as shown, the side piece 90 can include one or more contour features, such as ribs or fins. The side piece 90 can be connected to the rear drive assembly by a mechanical fastener and/or a snap-fit configuration. As also shown, the motor 93 can be contained in a protective shell, such as a generally cylindrical casing 97. In some embodiments, the side piece 90 and the casing 97 form a unitary and/or integral unit. In other embodiments, the cylindrical casing 97 is separate from the side piece 90.

A close-up partial exploded view of the rear drive assembly 80 is illustrated in FIG. 6. The side piece 90 is removed to illustrate how various components connect together to provide power to rear wheel 30. In some embodiments, the rear drive assembly 80 includes a gear 84 that connects to (e.g., is received in) a rotation connector 82. The gear 84 and connector 82 are received (e.g., wedged) within the rear wheel 30. In some configurations, this can provide a friction fit to secure the gear 84 and connector 82 within the rear wheel 30.

In some embodiments, the rotation connector 82 comprises a one-way bearing connector. The one-way bearing connector 82 can be configured to transmit torque in one direction and allow free motion in the opposite direction. In certain implementations, the one-way bearing connector 82 allows the wheel 30 to spin freely (also called “freewheeling”) in a one rotational direction and can be driven in an opposite rotational direction. For example, relative to the orientation of the scooter 10, the one-way bearing connector 82 can enable the wheel 30 to freewheel when the scooter 10 is moved backward and to be driven to move the scooter 10 forward. In some embodiments, the wheel 30 is configured to rotate about a wheel axis 88 and/or to be driven by the motor 93, such as through gearing. In some implementations, the one-way bearing 84 reduces or eliminates undesirable noises, such as clicking noises commonly heard with bearings, by allowing the rear wheel 30 to freewheel. The one-way bearing connector 82 fits within a connector receiving area 86 of a gear box housing 87. The gear box housing 87 can house and/or surround at least one or more components of the rear drive assembly 80. In some embodiments, the rear box housing 87 can connect directly and/or indirectly to the side piece 90 to surround at least a portion of the rear drive assembly 80. The rear drive assembly 80 can include a plurality of bearings 81, 83, and 85.

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In some embodiments, the rear drive assembly **80** includes a brake mechanism **96**. The brake mechanism **96** may be in the form of a lever, as shown in FIG. **6**, which can be activated by a user's foot to apply pressure to the rear wheel **30** and result in a braking force due to friction between the rear wheel **30** and the brake mechanism **96**. The brake mechanism **96** may be formed integrally with the main structure of the rear deck or body of the scooter **10** or may be a separate component coupled with the main structure of the rear deck or body of the scooter **10**. In some embodiments, the brake mechanism **96** is similar or identical to the brake assembly disclosed in U.S. Pat. No. 8,813,892, which is hereby incorporated by reference in its entirety. In some embodiments, the brake mechanism **96** comprises a rear end that is configured to engage with the rear wheel **30** and/or a front end that is configured to engage with the casing **97**. The brake mechanism **96** can be biased by a biasing member, such as a spring. For example, the rear end of the brake mechanism **96** can be biased away from the rear wheel **30** and/or the front end of the brake mechanism **96** can be biased into engagement with a stop structure, such as the casing **97**. As shown in FIG. **1**, for example, the brake mechanism **96** can extend rearwardly from the main structure of the rear deck or body of the scooter **10** and can extend over at least a portion of the rear wheel **30**.

FIG. **7** illustrates a view of the opposite or gear box side of the rear drive assembly **80** and FIG. **8** illustrates an exploded view of the side of the rear drive assembly **80** shown in FIG. **7**. A gear set **89** is housed within the gear box housing **87**. The gear box housing **87** may be part of the main structure of the rear deck or body of the scooter **10**. A portion of the gear set **89** can be enclosed on the outside with a gear box side piece **92** and a cover piece **91**. The side piece **92** can include one or more apertures, such as to allow airflow from ambient to the motor **93** for cooling. As shown, in some embodiments, the side piece comprises a stepped configuration, such as a first step **93a**, a second step **93b**, a third step **93c**, and a fourth step **93d**. Each of the steps can have a circular peripheral shape. In some embodiments, as illustrated, the first, second, and third steps **93a-93c** can have respective outer surfaces that are laterally offset (e.g., non-coplanar) from each other. In some implementations, the outer surfaces of the first and fourth steps **93a**, **93d** are substantially co-planar. In some embodiments, the second step **93b** includes at least one fin that extends forwardly from the second step **93b** to the first step **93a**. The gear set **89** can include a plurality of gears (e.g., a worm gear, spur gear, etc.) that can be used to convert the torque produced by the motor **93** into a torque that is used to power the rear wheel **30**. The motor **93** may be housed within the gear box **87** at or near the end of the deck or body of the scooter **10**. As illustrated, the gear box **87**, in some configurations, forms a part of the structure of the rear deck or tail of the scooter **10**. In some embodiments, the motor **93** may be a hub motor similar to those described in U.S. Patent Publication No. 2015/0133253, which is incorporated by reference herein in its entirety. In certain embodiments, the motor **93** may be a small toy motor, such as those used to drive electric motor-driven toys.

As shown in FIGS. **8-10**, the motor **93** drives a motor shaft **94**, such as a driving shaft. In the embodiment illustrated, a gear **95a** of the gear set **89** is mounted on the shaft **94** and a gear **95e** of the gear set **89** is mounted on the rear wheel axis **88**. The gears **95a**, **95e** can be operably connected through intermediate gears, such as gears **95b**, **95c**, **95d**. Through the sequence of gears **95a-95e**, the torque of the motor **93** is reduced to drive the rear wheel via the rear wheel

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axis **88**. As discussed above, the gear **84** and bearing connector **82** allow the rear wheel **30** to be powered in a first, or forward direction, and allow the rear wheel **30** to freewheel in a second, or rearward direction. An exploded view of the rear drive assembly **80** is illustrated in FIGS. **9** and **10**. In various embodiments, the rear wheel **30** can freewheel with no or substantially no sound (e.g., clicking) that is detectable by a user in normal operation of the scooter. For example, in some embodiments, during freewheeling, the wheel **30** and/or the connector **82** emits a sound that is less than or equal to about 20 decibels.

In some embodiments, the gear **84** has a diameter that is wide enough to be received within a receiving area of the bearing connector **82** and within the receiving area **86** of the gear box housing **87**, as can be seen in at least FIG. **6**. The gear **84** has a plurality of flanges **84a** extending radially outward from the outer surface of the gear **84**. The flanges **84a** are configured to fit within corresponding recesses **84b** located on the inside surface of the bearing connector **82**. Similarly, the inside surface of the receiving area **86** of the gear box housing has recesses **84c** that are also configured to receive the flanges **84a**. The length **L** of the gear **84** is larger than the length **L'** of the bearing connector **82** such that at least a portion of the gear **84** is received within the bearing connector **82** and at least a portion of the gear **84** is received within the receiving area **86** of the gear box housing **87**.

Rotation of the rear wheel axis **88** acts on the bearings **83**, **85**, which may be press fit or otherwise coupled to the axis **88**. The bearings **83**, **85** can be coupled with the gear **84** through a friction fit or other coupling mechanism such that rotation of the axis **88** also rotates the gear **84**. Rotation of the gear **84** in a first rotational direction induces rotation of the bearing connector **82** through the interaction of the flanges **84a** on the recesses **84b** and rotates the rear wheel **30** in the first rotational or driven direction. As discussed above, the bearing connector **82** also permits freewheel rotation of the rear wheel **30** in a second rotational direction opposite the first rotational direction when the motor **93** is not powering the rear wheel **30** through the rear drive assembly **80**, as discussed above.

FIGS. **11-13** illustrate the scooter **10** in a folded configuration. As discussed above, the folding assembly **70** allows the handlebar assembly **24** to pivot relative to the body **20** such that the rotatable shaft **32** and the handlebar assembly **24** fold against to the body **20** for compact transport and storage of the scooter **10**. As shown in FIGS. **11-13**, when the scooter **10** is in the folded configuration, the handlebar assembly **24** can be positioned generally parallel to the body **20**. The folding assembly **70** includes a release mechanism **72**. The release mechanism **72** can include a handle, button, lever, and/or knob, among mechanical mechanisms. For example, in the illustrated embodiment, the release mechanism **72** includes a handle that, in a first position, locks the rotatable shaft **32** in a substantially upright or riding position and in a second position, allows the rotatable shaft to pivot about a pivot connection **74** (shown in FIG. **13**) between the rotatable shaft **32** and the body **20** such that the rotatable shaft **32** is substantially parallel with a longitudinal axis of the body **20**, as shown in FIG. **11**. In some embodiments, the release mechanism **72** is in wired and/or wireless communication with the controller to receive instructions to release and/or lock the folding assembly **70**. In some embodiments, when the scooter **10** is in the folded configuration, the handlebar assembly is positioned adjacent or near the brake mechanism **96**. In some embodiments, the handlebar assembly **24** abuts and/or touches the brake mechanism **96** and can

cause the brake mechanism **96** to contact the rear wheel **30** or otherwise lock the rear wheel **30** to prevent rotation of the rear wheel **30**.

Some embodiments of the electric scooter **10** comprises a battery housing **22** comprising a controller and at least one battery **21**. As shown, the housing **22** can be mounted to the underside of the scooter body **20** underneath the support surface **34**. As illustrated in FIGS. **11** and **12**, the battery **21** is insertable and/or removable from the battery housing **22** when the scooter **10** is in a folded configuration. The battery **21** may be any type of battery, such as a lithium ion rechargeable battery. For example, the battery can have an approximate 1.5-2.5 hour discharge time. As illustrated, the battery **21** can be inserted and/or removed along a direction that is generally parallel with a longitudinal axis of the body **20**.

FIG. **13** illustrates a cross-sectional view of the scooter **10** in the folded configuration. As shown, the battery housing **22** can be located at the forward end of the body **20** of the scooter **10**. This arrangement facilitates easy removable and replacement of the battery **21** when the scooter is in a folded configuration, allowing access to the battery housing **22**. The battery housing can include an aperture on the forward end of the body **20**. The housing **22** can include a closure feature, such as a door (not shown), to close the aperture. In some embodiments, the body **20** and/or the battery **21** comprise a securing feature, such as a detent or hook, configured to secure the battery within the housing **22**. The securing feature can include a release, such as a button or latch, that disengages the securing feature and/or enables the battery **21** to be removed from the housing **22**.

The battery **21** can be configured to be moved into and out of the housing **22**. For example, the battery **21** can be slid into and out of the housing **22** along an axis that is generally parallel with a longitudinal axis of the body **20**. In certain implementations, the battery **21**, when received in the housing **22**, is positioned forward of a front to rear midpoint of the body **20**. This can aid in moving the center of gravity of the scooter forward, which can provide advantages in riding characteristics. In some variants, the battery **21** is configured to be received in a cavity in a rear of the scooter, such as near the rear wheel **30**. This can increase the amount of weight that is distributed to the rear wheel, which can increase traction of the rear wheel **30**.

The body **20** can include a middle connection piece **23** to provide a point of connection between the support surface **34** and the body **20**, as well as provide additional stability and support. A controller **25** can be located rearward of the battery **21** of the body **20** below the support surface **34**. In some embodiments, the controller **25** is close to the motor **93** and/or between the motor **93** and the battery **21**. In some embodiments, during assembly, the controller **25** may be coupled to the middle connection piece **23**, such as by a clip-in connection, friction fit, fasteners, or otherwise.

As discussed above, some embodiments comprises the motor **93**, such as an electric motor. The motor **93** can be mounted to the underside of the scooter body **20** underneath the support surface **34**, mounted on a rear end of the support surface **34** adjacent the rear drive assembly **80**, mounted on a front end of the support surface **34** adjacent the front wheel **28**, mounted on the handlebar assembly **24** or fork **31**, and/or in any other location. The motor can be operably coupled to the at least one rear wheel **30** to provide powered rotational force upon the at least one rear wheel **30**. In some embodiments, the battery housing **22** and the motor **93** can be located or supported at different locations on the scooter **10**,

including on top of the support surface **34** or adjacent either the front or rear wheel **28**, **30**.

In various embodiments, the controller **25** can receive a signal from the throttle assembly **26**. For example, the controller **25** can receive a signal indicative of the amount of speed and/or power to apply to the rear wheel **30**. The controller **25** can provide two-way or one-way transmission to the motor **93**. For example, the controller **25** can instruct the motor **93** to drive the wheel **30** in response to and/or consistent with the signal from the throttle assembly **26**. While control of the scooter **10** can be wireless via the wireless throttle assembly **26**, some variants have wired connections may be provided to connect the throttle, brake, and on/off switch to the motor **93**. Any wired or wireless protocol may be used.

In some embodiments, the scooter **10** includes no wires connecting the battery **21**, controller **25**, and motor **93**. In some embodiments, a conductive material, such as copper, may be incorporated into the body **20** of the scooter **10** once the body **20** has been formed (e.g., injection molded). The conductive material running through the injection molded body **20** acts as a conductor and electrically connects the battery **21**, motor **93**, and controller **25** to control driving operation of the rear wheel **30**. The rear drive assembly **80** may, in some embodiments, be a single module or assembly with the controller and battery separate modules that may be coupled to or inserted within the body **20** during manufacture.

FIGS. **15-21**

FIGS. **15-21** illustrate another embodiment of a scooter **110**. The scooter **110** is similar or identical to the scooter **10** discussed above in many respects. As shown in FIGS. **15-21**, the scooter **110** can include a scooter body **120**, a support surface **134**, a rotatable shaft **132**, and a handlebar assembly **124**, which can be similar to the scooter body, the support surface, the rotatable shaft, and the handlebar assembly discussed above in connection with the scooter **10**. The scooter **110** can include any one, or any combination, of the features of the scooter **10**.

For example, the scooter **110** includes at least one front wheel **128** and at least one rear wheel **130** connected with the scooter body **120**. The front wheel **128** and the rear wheel **130** can be spaced apart from one another with the support surface **134** extending therebetween. The wheels **128**, **130** can be aligned in substantially the same plane, as depicted, and/or located at opposite ends of the support surface **134**. As depicted, some embodiments of the scooter **110** include a rotatable shaft **132** which rotates with the handlebar assembly **124** so that the handlebar assembly **124** can rotate or swivel within a head tube of the scooter body **120**.

In some embodiments, the scooter **110** includes a folding assembly **170** and/or a motor assembly, such as a rear drive assembly **180**. The folding assembly **170** allows the handlebar assembly **124** and the rotatable shaft **132** to pivot relative to the body **120** such that the handlebar assembly **124** folds against the body **120** for compact transport and storage of the scooter **110**. As shown in FIGS. **15-21**, when the scooter **110** is in the folded configuration, the handlebar assembly **124** can be positioned generally parallel to the body **120**. The folding assembly **170** includes a release mechanism **172**. The release mechanism **172** includes a knob that, in a first position, locks the rotatable shaft **32** in a substantially upright or riding position and in a second position, allows the rotatable shaft to pivot about a pivot connection **174** between the rotatable shaft **132** and the body **120** such that the rotatable shaft **132** is substantially parallel with a lon-

gitudinal axis of the body **120**. For example, the release mechanism **172** can be rotated in a first direction to lock the rotatable shaft **132** and a second direction, opposite the first direction, to release the rotatable shaft **132** and allow the rotatable shaft **132** to pivot about the pivot connection **174**.

In some embodiments, the rear drive assembly **180** includes a gear box housing **187**, which can at least partially enclose a gear set. The gear box housing **187** may be part of the main structure of the rear deck or body of the scooter **110**. A portion of the gear set **189** can be enclosed on the outside with a gear box side piece **192**. As shown, the side piece **192** can include a stepped configuration. For example, the stepped configuration can include a first portion **192a** and a second portion **192b**. As shown in the illustrated configuration, the first portion **192a** is substantially flat and the second portion **192b** includes a stepped section that protrudes outwardly from first portion **192a** and is laterally offset from the first portion **192a**. The stepped section can be positioned at least partially adjacent the motor and can extend a portion of the first portion **192a**. The side piece **192** can include one or more apertures **191**, such as to allow airflow from ambient to the motor **193** for cooling. The apertures **191** can be positioned between the first portion **192a** and the second portion **192b** and at least partially surround an end of the motor **193** to allow airflow to directly contact the motor **193** for cooling.

In some embodiments, the rear drive assembly **180** includes a side piece **190** that is removable to provide access to the rear wheel **130** in case the rear wheel should need repair or replacement, such as due to wear, etc. The side piece **190** can be connected to the rear drive assembly **180** to surround at least a portion of the rear drive assembly **180** and can be positioned opposite the side piece **192**, with the gear box housing **187** extending therebetween. As shown in the illustrated embodiment, the side piece **190** can be substantially flat.

As also shown, the motor **193** can be contained in a protective shell, such as a generally cylindrical casing **197**. In some embodiments, the side piece **190** and the casing **197** form a unitary and/or integral unit. In other embodiments, the casing **197** is separate from the side piece **190**. In some embodiments, the casing **197** is formed as part of the motor **193** and includes one or more apertures to allow airflow from ambient to the motor **193** for cooling.

Certain Terminology

Terms of orientation used herein, such as “top,” “bottom,” “horizontal,” “vertical,” “longitudinal,” “lateral,” and “end” are used in the context of the illustrated embodiment. However, the present disclosure should not be limited to the illustrated orientation. Indeed, other orientations are possible and are within the scope of this disclosure. Terms relating to circular shapes as used herein, such as diameter or radius, should be understood not to require perfect circular structures, but rather should be applied to any suitable structure with a cross-sectional region that can be measured from side-to-side. Terms relating to shapes generally, such as “circular” or “cylindrical” or “semi-circular” or “semi-cylindrical” or any related or similar terms, are not required to conform strictly to the mathematical definitions of circles or cylinders or other structures, but can encompass structures that are reasonably close approximations.

Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include or do not

include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Conjunctive language, such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

The terms “approximately,” “about,” and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, in some embodiments, as the context may permit, the terms “approximately,” “about,” and “substantially” may refer to an amount that is within less than or equal to 10% of the stated amount. The term “generally” as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. As an example, in certain embodiments, as the context may permit, the term “generally parallel” can refer to something that departs from exactly parallel by less than or equal to 20 degrees.

Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B, and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Likewise, the terms “some,” “certain,” and the like are synonymous and are used in an open-ended fashion. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

Overall, the language of the claims is to be interpreted broadly based on the language employed in the claims. The language of the claims is not to be limited to the non-exclusive embodiments and examples that are illustrated and described in this disclosure, or that are discussed during the prosecution of the application.

SUMMARY

Although this invention has been disclosed in the context of certain embodiments and examples, it will be understood by those skilled in the art that the present disclosure extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. In particular, while the present systems and methods have been described in the context of particular embodiments, the skilled artisan will appreciate, in view of the present disclosure, that certain advantages, features and aspects of the systems and methods may be realized in a variety of other applications, many of which have been noted above. Various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes

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of the conveyor. The scope of this disclosure should not be limited by the particular disclosed embodiments described herein.

Additionally, it is contemplated that various aspects and features of the invention described can be practiced separately, combined together, or substituted for one another, and that a variety of combination and subcombinations of the features and aspects can be made and still fall within the scope of the invention. Certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as any subcombination or variation of any subcombination.

Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, and all operations need not be performed, to achieve the desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products. Additionally, other implementations are within the scope of this disclosure.

Some embodiments have been described in connection with the accompanying drawings. The figures are drawn to scale, but such scale should not be limiting, since dimensions and proportions other than what are shown are contemplated and are within the scope of the disclosed invention. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various embodiments can be used in all other embodiments set forth herein. Additionally, any methods described herein may be practiced using any device suitable for performing the recited steps.

In summary, various embodiments and examples of personal mobility devices, such as scooters, have been disclosed. Although the devices have been disclosed in the context of those embodiments and examples, this disclosure extends beyond the specifically disclosed embodiments to other alternative embodiments and/or other uses of the embodiments, as well as to certain modifications and equivalents thereof. This disclosure expressly contemplates that various features and aspects of the disclosed embodiments can be combined with, or substituted for, one another. Thus, the scope of this disclosure should not be limited by the particular embodiments described above, but should be determined only by a fair reading of the claims that follow.

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What is claimed is:

1. A two-wheel scooter comprising:

- a body comprising a deck, a front end, a rear end, and a longitudinal axis, the deck configured to support a user;
- a steering assembly connected to the front end of the body, the steering assembly comprising a handlebar, a steering shaft, and a front wheel;
- a rear wheel connected to the rear end of the body;
- a rear drive assembly configured to drive the rear wheel, the rear drive assembly comprising an electric motor and a transmission;
- a battery housing under the deck, the battery housing comprising an aperture; and
- a battery in the battery housing, the battery configured to provide electric power to the electric motor, the battery selectively removable from the battery housing through the aperture along a direction that is generally parallel with the longitudinal axis of the body, wherein when the battery is in the battery housing a front end of the battery protrudes longitudinally frontward from the front end of the body, thereby providing a grasping portion to aid removal of the battery.

2. The scooter of claim 1, further comprising a folding assembly that connects the steering assembly and the body, the folding assembly comprising a hinge and a lock, the folding assembly configured to enable the steering assembly to pivot relative to the body between an unfolded state and a folded state, wherein:

- in the unfolded state, the steering assembly blocks removal of the battery from the battery housing; and
- in the folded state, the steering assembly does not block removal of the battery from the battery housing.

3. The scooter of claim 2, wherein, in the unfolded state and with the front and rear wheels on a riding surface, the battery is directly below the folding assembly.

4. The scooter of claim 1, wherein the direction is toward the front end of the body.

5. The scooter of claim 1, wherein the aperture is on a front end of the battery housing.

6. The scooter of claim 1, wherein the battery housing is connected to the front end of the body.

7. The scooter of claim 1, wherein the rear drive assembly is configured to drive the rear wheel in a first rotational direction and is configured to allow the rear wheel to rotate freely in a second rotational direction opposite of the first rotational direction.

8. A method of replacing a spent battery in an electric scooter, the method comprising:

- disengaging a lock of a folding assembly of the scooter; folding the scooter, wherein folding the scooter comprises:
 - pivoting in a first rotational direction the steering assembly of the scooter relative to a body of the scooter; and
 - activating, with the steering assembly, a brake of the scooter against a rear wheel of the scooter;
- withdrawing the spent battery from the body in a first linear direction generally parallel to a longitudinal axis of the body;
- inserting a fresh battery into the body in a second linear direction generally parallel to the longitudinal axis;
- unfolding the scooter, wherein unfolding the scooter comprises pivoting in a second rotational direction the steering assembly relative to the body; and
- engaging the lock of the folding assembly.

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9. The method of claim 8, wherein the first linear direction is toward a front of the scooter and the second linear direction is toward a rear of the scooter.

10. The method of claim 8, wherein folding the scooter further comprises moving a front wheel of the scooter out of a position that blocks withdrawal of the spent battery, and wherein unfolding the scooter further comprises moving the front wheel into a position that blocks withdrawal of the fresh battery.

11. The method of claim 8, wherein withdrawing the spent battery from the body comprises grasping a front of the battery that protrudes from the body of the scooter.

12. A two-wheel scooter comprising:

a body comprising a deck, a front end, a rear end, and a longitudinal axis, the deck configured to support a user; a steering assembly connected to the front end of the body, the steering assembly comprising a handlebar, a steering shaft, and a front wheel;

a rear wheel connected to the rear end of the body;

a rear drive assembly configured to drive the rear wheel, the rear drive assembly comprising an electric motor and a transmission;

a battery configured to provide electric power to the electric motor;

a brake configured to move into contact with the rear wheel; and

a folding assembly configured to enable the steering assembly to pivot relative to the body between an unfolded state and a folded state, wherein in the folded state the steering assembly contacts the brake, thereby inhibiting rotation of the rear wheel.

13. The scooter of claim 12, wherein the folding assembly comprises a hinge and a lock.

14. The scooter of claim 13, wherein the lock comprises a rotatable knob.

15. The scooter of claim 12, wherein the battery is selectively removable from the body along a direction that is generally parallel with the longitudinal axis of the body.

16. The scooter of claim 12, wherein the brake is configured to be activated by a rider's foot.

17. The scooter of claim 12, wherein the rear drive assembly is configured to drive the rear wheel in a first

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rotational direction and is configured to allow the rear wheel to rotate freely in a second rotational direction opposite of the first rotational direction.

18. A two-wheel scooter comprising:

a body comprising a deck, a front end, a rear end, and a longitudinal axis, the deck configured to support a user; a steering assembly connected to the front end of the body, the steering assembly comprising a handlebar, a steering shaft, and a front wheel;

a rear wheel connected to the rear end of the body;

a rear drive assembly configured to drive the rear wheel, the rear drive assembly comprising an electric motor and a transmission;

a battery housing under the deck, the battery housing comprising an aperture;

a battery in the battery housing, the battery configured to provide electric power to the electric motor, the battery selectively removable from the battery housing through the aperture along a direction that is generally parallel with the longitudinal axis of the body; and

a folding assembly that connects the steering assembly and the body, the folding assembly comprising a hinge and a lock, the folding assembly configured to enable the steering assembly to pivot relative to the body between an unfolded state and a folded state, wherein: in the unfolded state and with the front and rear wheels on a riding surface, the steering assembly blocks removal of the battery from the battery housing and the battery is directly below the folding assembly; and

in the folded state, the steering assembly does not block removal of the battery from the battery housing.

19. The scooter of claim 18, wherein the direction is toward the front end of the body.

20. The scooter of claim 18, wherein the aperture is on a front end of the battery housing.

21. The scooter of claim 18, wherein the battery housing is connected to the front end of the body.

22. The scooter of claim 18, wherein the rear drive assembly is configured to drive the rear wheel in a first rotational direction and is configured to allow the rear wheel to rotate freely in a second rotational direction opposite of the first rotational direction.

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